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MILITARY STANDARD

TESTS FOR

CONSTRUCTION, INDUSTRIAL, AND MATERIALS HANDLING EQUIPMENT

FSC MISC

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DEPARTMENT OF DEFENSE

WASHINGTON, DC 20301

Tests for Construction, Industrial
and Materials Handling Equipment

MIL-STD-448A

1. This Military Standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Recommended corrections, additions, or deletions should be addressed to the Commander, U.S. Army Mobility Equipment Research and Development Center, Fort Belvoir, VA 22060.

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FOREWORD

This standard was originally developed by the Department of Defense in conjunction with industry when the need for standard test methods for construction and industrial machines became evident. Since publication, advancement in complexity and capabilities of materials handling equipment has reached a point where it is feasable to include this classification of equipment in this standard. This standard is based on trade practices and terminology and is intended to be used in determining the capabilities of individual equipment to meet the requirements of the end item specifications.

Part I contains general information, establishes criteria on test equipment, establishes test conditions, defines engineering terms used in the different tests, and establishes controls on variables necessary to repeat tests with reasonably consistant results.

Part II contains specific individual recommended tests.

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CONTENTS

PART I

TEST CONDITIONS AND APPARATUS

Section	Content	Page
1	SCOPE	1
1.1	Scope	1
1.2	Application	1
1.3	Form and Numbering	1
1.4	Numbering System	1
2	REFERENCE DOCUMENTS	2
3	DEFINITIONS	3
4	GENERAL TEST CONDITIONS	17
4.2	Fuels, Lubricants, and Coolants	17
4.3	Operating Temperatures	17
4.4	Preparation of Test Item	17
5	TEST COURSES	17
5.2	Level Tractive Ability Test Course	17
5.3	Test Course Surfaces	17
6	TEST APPARATUS	19
6.1	Scope	19
6.2	Application	19 and 20
6.3	Acceleration	20
6.4	Fluid Volume	21
6.5	Force	21
6.6	Gradient	22
6.7	Linear Dimensions	22 and 23
6.8	Particle Counting	23
6.9	Pressure	23 and 24
6.10	Specific Gravity	24
6.11	Speed	24 and 25
6.12	Stress	25
6.13	Temperature	25, 26, and 27
6.14	Time	27
6.15	Weight	28
6.16	Wind Direction and Velocity	28

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1. SCOPE

1.1 Scope. This standard establishes tests that apply to the different types of construction, industrial, materials handling, and the auxiliary equipment and attachments thereto. It sets forth ways of and means for determining the performance ability and physical aspects of machinery and eliminates unnecessary and undesirable variations in tests. This standard does not cover all of the tests used in determining acceptability of different equipment nor does it attempt to cover all of the special tests applicable to individual special purpose equipment.

1.2 Application. The provisions of this standard apply to the following:

(a) End item specifications.
(b) Doctrinal criteria for equipment mission suitability.

1.3 Form and numbering. The section contains a brief description and manner in which new tests are added, old tests revised, and individual tests numbered.

1.3.1 Form. A loose leaf form is used for convenience in inserting or removing the material needed to revise the contents and thus keep abreast of latest testing techniques. Suggestions for improvement of technical content are solicited from users.

1.3.2 New tests. New tests will be assigned an appropriate number within a particular group of tests. For example, a new test for the determination of the travel speed of a mobile crane is properly placed in the end item tests.

1.3.3 Revision of existing methods. When changes or corrections are necessary, the method will be rewritten and given a decimal number and assigned an effective date. When the revision is promulgated, it shall automatically supersede all previous editions; and shall be used whenever the test is referenced in product specifications.

1.4 Numbering system. All tests in this standard are originally identified by a group of three digits, which is comprised of two basic parts to signify the following:

(a) First digit - category of test.
(b) Second and third digits - test number within a category of tests.

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1.4.1 Category number classification. The first digit of a test number denotes the category of the test. These categories are as follows:

Digit	Category
1	End Item Test - A test which examines the mobility, performance, reliability, transportability, maintainability, operability, safety, or other function of an end item.
2	System Test - A test which examines such things as the performance, reliability, and maintainability of a system which forms part of an end item. These may include the electrical, hydraulic, fuel, cooling, drive train, power module and braking system of the end item.
3	Component Test - A test which examines such things as the performance, reliability, or maintainability of a component which forms part of a system included in the end item. These may include hydraulic, electrical, power module and drive train components.

1.4.2 Test number. The number of a test under a category of tests.

1.4.3 Example of number system.

Test	189
Category - End item test	1
Test number of series	89

2. REFERENCE DOCUMENTS

2.1 Specifications and standards. (Not applicable)

2.2 Other publications. The following documents form a part of this standard to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D1883 - Bearing Ratio of Laboratory-Compacted Soils.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B93.20 - Procedure for Qualifying and Controlling Cleaning Methods for Hydraulic Fluid Power Fluid Sample Containers.
B93.28 - Method for Calibration of Liquid Automatic Particle Counters Using "AC" Fine Test Dust.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

3. DEFINITIONS

3.1 Definitions. The following definitions apply to the terms and phrases used in this standard.

A

Accessory: A supplementary device designed for use with an end-item, and affixed either permanently or semipermanently thereto, contributing to the effectiveness thereof without extending or varying the basic function of the end-item. Sometimes referred to as auxiliaries.

Attachment: A supplementary device designed to be affixed to, or mounted on, an end-item in order to extend or vary the function of the end-item.

B

Brake systems:

1. Auxiliary brake - a brake, independent in application from the service brake, used in parking and controlling the vehicle when service brake fails, sometimes called a parking or handbrake; a locking device is provided to keep the brake applied while the vehicle is unattended.

2. Brake lag distance - the distance traveled between the point where the driver begins to move the brake controls and the point of first retardation.

3. Braking efficiency - the ratio of the stopping effort and the weight of a vehicle, either with or without ballast or payload as specified.

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4. Braking effort - the frictional force available to resist rotation of each wheel of the vehicle with the brake fully applied.

5. Driver reaction distance - the distance traveled by a vehicle between the point where the driver perceives the need for braking and the point where he starts to move the brake controls.

6. Rolling radius - the distance from the center of the wheel axle to the ground level at a specified tire pressure and a given weight on the wheel.

7. Service brake - primary brake for retarding, stopping, and controlling the vehicle.

8. Stopping effort - average retardation force applied through the brake system during travel over stopping distance.

9. Travel speed - the speed of vehicle at the moment the driver begins moving the brake controls.

10. Vehicle braking distance - the distance the vehicle travels between the point of first retardation and the point where the vehicle comes to rest.

11. Vehicle stopping distance - the distance traveled between the point where the driver activates the brake system and the point where the vehicle comes to rest.

C

Component: A constituent part of a system or end-item without which the end-item is incomplete.

Conventional draft gears: The typical coupling system used on railroad cars. Cushion couplers are not to be included.

Crane Axis-of-Rotation: A vertical line through the axis around which the crane superstructure rotates, before load is applied to the crane hook.

Crane Balance-point: The condition of crane loading wherein the load moment acting to overturn the crane is equal to the maximum moment of the crane available to resist overturning; this is observed as that point at which the wheels or outriggers start to lift off the surface.

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Crane Hoisting force: Hoist-line pull, in pounds, require to lift a specified load.

Crane Hoist-line: A wire rope or similar line used with sheaves and a drum to lift and support loads.

Crane Hoist-line power: The rate of work done by the hoist-line, determined from measured values of force, distance, and time, normally expressed as horsepower or foot-pounds per unit of time.

Crane Load: The force acting to unbalance a crane; it results from (1) the gravitational force created by all items suspended from the hook block; (2) force exerted by hoisting on a fixed anchor; or (3) a combination of the above forces.

Crane Radius-of-Load: The horizontal distance from a projection of the axis-of-rotation to the supporting surface, before loading, to the center of vertical hoist line or tackle with load applied.

Cylinder port: A port which provides a passage to or from an actuator.

D

Dimensions: All measurements of physical dimensions shall be made with the equipment on a level surface with zero surface penetration, unless otherwise specified in the test method.

1. Overall length - the horizontal projected distance from the most forward part of the end-item, including bumpers, lights, grills, etc., to the most rearward part, with the accessories or attachments in normal traveling position.

2. Overall width - the horizontal projected distance from the farthest projecting part of one side of the end-item to the farthest projecting part of the other side with the accessory or movable attachment in normal traveling position. In measuring:

(a) dozers: the dozer shall be at right angles to the longitudinal axis of the vehicle or machine; and
(b) graders: the blades shall be in the traveling position - if tire deflection or bulge is a factor, the tire load and inflation pressure must be specified.
(c) trucks, lift, fork: the forks shall be in the traveling position.

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3. Overall height - the vertical distance from the ground level at zero surface penetration to the highest projection with accessories or attachments in traveling position.

Directional control valve: A valve whose primary function is to direct or prevent flow through selected passages.

Drag forces:

1. Vehicle drag - resistance to travel (horizontal force coinciding with direction of travel) over a specified test course, the surface wind being at zero; resistance includes but may not be limited to chassis friction, air resistance, friction in tire walls and treads, deformation of travel surface, and impact resistance from irregularities of road surface.

2. Wind resistance - force acting on a vehicle during travel. The angle of wind force to the direction of travel shall be stated.

Drawn vehicles: Vehicles attached to the prime mover by means of a drawbar hitch or some other device designed for the purpose of towing or pulling.

E

End-item: A combination of components, systems, parts and assemblies which make up an equipment, complete and ready to serve its intended function.

G

Gear selections:

Overall gear reduction - the step-down gear ratio between the engine (flywheel) and the drive wheel or drive sprocket.

H

Hammer car: The car which is accelerated to a speed and humped into the five buffer cars. This car usually has the test item loaded on it.

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Horsepower:

1. Observed reserve tractive - power ability of vehicle, as observed, to perform useful work beyond that required to move itself.

2. Corrected reserve tractive - observed tractive horsepower corrected to establish standard conditions.

Hydraulic horsepower:

Hydraulic Horsepower = $5.8 \times 10^{-4} \times Q \times P$
where: Q = flow in gpm, P = pressure in psi.

Hydraulic leakage: Flow from the tank port of the component through the sealing surfaces at some pressure differential.

M

Maintenance:

1. Preventive maintenance - the systematic care, servicing and inspection of equipment for the purpose of keeping it in serviceable condition and detecting and correcting incipient failures; it consists primarily of inspection, lubrication, adjustments, cleaning, minor parts replacement, tune-up, miscellaneous.

2. Inspection - periodic examination of machine components and oil, lubricant and coolant levels, etc., as specified in maintenance manuals; also examination on occasions when the equipment or any of its components is not functioning properly.

3. Lubrication - lubrication of bearings, linkages, etc.; draining and replacement of lubricants in the engine crankcase, gear, clutch, pump, final drive housings, etc.

4. Cleaning - periodic cleaning or flushing of cooling and lubrication systems of engines, gear and clutch housings, oil and fuel filters, battery terminals, dirt residue from engine, and mechanical parts, etc.

5. Adjustments - periodic check of the various equipment components and the necessary periodic adjustments specified in the maintenance manuals to insure effective and efficient performance. This includes battery charging, spark plugs, breaker points, injectors, fuel pump pressures, braking systems, gear shifting mechanisms, steering clutches, and gears, engine valve clearances, etc.

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6. Minor parts replacement - this varies, depending on the type of equipment and its use; parts in this category are specified in the maintenance manuals; they include fuel and oil filters, spark plugs, breaker points, belts, generator and motor brushes, fuel injector and carburetor parts, etc., which are normally inspected and serviced periodically.

7. Tune-up - applies to the power plant equipment such as electric generators, and starting engines or motors and includes the cleaning of parts, checking of valve clearances, operation of the timing and fuel supply systems, and the performance of specified adjustments, part replacements, and lubrication. (If part replacements and adjustments are major operations requiring special skills, the work is considered corrective rather than preventive maintenance.)

8. Corrective maintenance - maintenance and repair required over and above preventive maintenance wherein items that are unserviceable are made serviceable by the adjustment or replacement of damaged or unserviceable parts, components, or assemblies; this normally is comparable with field maintenance; it may also include work to rebuild and restore the item to a condition comparable with new by disassembling its components to determine their condition and reassembling them, using serviceable, rebuilt or new assemblies, subassemblies, and parts; the latter is comparable with depot maintenance.

O

Operating clearance:

1. Ground clearance - a ground clearance is the perpendicular distance from the ground line to the lowest point under the center portion of the vehicle, measured from the midpoint of the tread gage; the center portion of the vehicle shall be 25 percent of the width of the tread gage to either side for track-type vehicles and 40 percent, for wheeled vehicles.

2. Drawbar height - the perpendicular distance from the ground line to a point midway between the jaws of the drawbar.

3. Drawbar swing angle - the maximum lateral angle of swing of the drawbar about its pivot point.

4. Angle of approach (see Fig. 1) - the measured angle between the horizontal and a plane; with the plane tangent to the forward tires or tracks of a vehicle and coincident with the lowest point of any structure or component forward of the tires or tracks, which limits the magnitude of the angle. Measurements are made with the vehicle on the horizontal and serviced or loaded to the degree specified for the determination.

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5. Angle of departure, (see Fig. 1) - the measured angle between the horizontal and a plane; with the plane tangent to the rear tires or tracks of a vehicle and coincident with the lowest point of any structure or component rear of the tires or tracks, which limits the magnitude of the angle. Measurements are made with the vehicle on the horizontal and serviced or loaded to the degree specified for the determination.

6. Interaxle angle of interference (see Fig. 1) - with all wheels in the same plane, it is the maximum possible angle between two planes intersecting on a downward projection of the vehicle, where the first plane is tangent to the wheels of the axle forward of the intersection and the second plane is tangent to the wheels of the axle behind the intersection.

7. Intertrack angle of interference (see Fig 2) - with both tracks in the same plane, it is the maximum possible angle between two planes intersecting on a downward projection of the vehicle, where the first plane is coincident with the inner edge of the left side track and the second plane is coincident with the inner edge of the right side track.

8. Interwheel angle of interference (see Fig. 3) - with all wheels of any axle resting in the same plane, it is the maximum possible angle between two planes intersecting on a downward projection of the vehicle where the first plane is tangent to the inside tire on the left side of the axle and the second plane is tangent to the inside tire on the right side of the axle.

Operating pressure: The pressure at which a system is operated (system relief valve pressure setting).

Overall pump efficiency: The hydraulic horsepower output of the pump divided by the measured mechanical input.

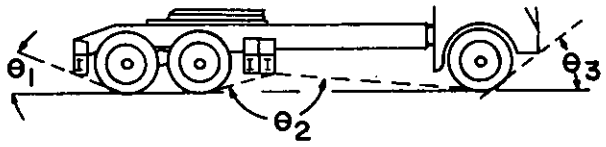
P

Particle counter branch line: Fluid conducting line teed into the bypass line and directing fluid into the counter sensor.

Particle counter bypass line: Fluid conducting line from the item being tested to the particle counter sensor and returns fluid to the item without passing through the sensor.

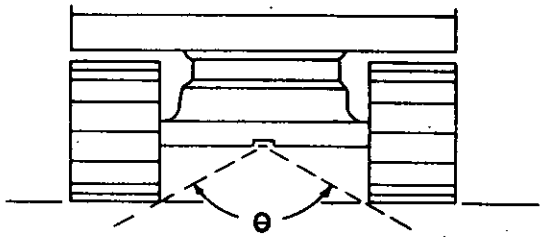
Particle counter saturation: The point at which the counter does not give a linear increase in counts with a linear increase in concentration of particles in the fluid being used.

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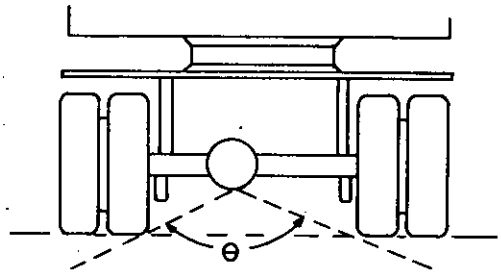
θ_1 - ANGLE OF DEPARTURE
θ_2 - INTER-AXLE ANGLE OF INTERFERENCE
θ_3 - ANGLE OF APPROACH

FIGURE 1.



θ - INTER - TRACK ANGLE OF INTERFERENCE

FIGURE 2.



θ - INTER - WHEEL ANGLE OF INTERFERENCE

FIGURE 3.

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P (Cont'd)

Piston drift: Gradual sinking of a piston under load over a period of time.

Pressure drop: The difference in pressure between any two points of a system or component.

Pressure port: A port which provides a passage from the source.

Proof pressure: The nondestructive test pressure, in excess of the maximum rated operating pressure, which causes no permanent deformation, excessive external leakage, or other malfunction.

Provision: An integral fitting or part of an end item, commonly called a padeye; lug, point, attachment, or fixture which may include a shackle or ring as an integral part. A provision affords a means of attaching a shackle or hook to equipment for tiedown or lifting during shipment.

S

Slave cylinder: Mechanical device that is directly responsive to another.

Specified: The term specified, where used herein, is construed to mean the recommendation of the manufacturer, the user, the test agency or any agreement between these parties. This includes requirements or conditions stated in the specifications or procurement documents.

Speed:

1. Nominal travel speed - the ground speed of equipment as observed by a visual indicator.

2. Measured travel speed - the average speed determined from the measured travel distance and travel time or the average speed determined from a continuous speed record.

System: A group of components which when put together perform a separate function on the end item.

T

Tractive effort:

1. Reserve tractive effort - the force developed by a vehicle beyond that required to move itself.

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2. Corrected tractive effort - reserve tractive effort corrected to establish standard conditions.

Tracks:

1. Track gage - the horizontal distance from the center of the drive sprocket on one side to the center of the teeth on the other side.

2. Surface penetration - the permanent vertical indentation of the travel surface due to one pass of one or more supporting wheels, tracks or skids.

3. Track idler adjustment - the distance the track idler on a vehicle may be moved forward or rearward from its midposition.

V

Valve cracking pressure: The pressure at which the pressure operated valve begins to pass fluid.

Valve reseat pressure: The pressure at which a pressure operated valve closes, restricting the flow of fluid.

W

Weights:

1. Ballast - a permanent or semipermanent load made a part of or added to the weight of a vehicle to increase its tractive ability and stability.

2. Center of gravity - the centroid of the forces of gravitation; its location is given by three dimensions, in a different plane, from specified reference planes or lines.

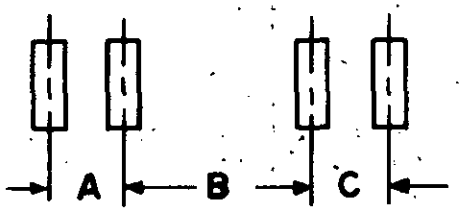
3. Net weight - weight of the vehicle without fuel, engine oil, coolant, on vehicle materiel, cargo, or operating personnel.

4. Payload - the cargo carried exclusive of driver.

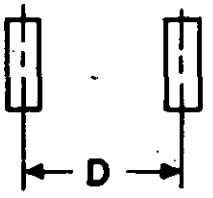
5. Curb weight - weight of the vehicle including fuel, lubricant, coolant, and on vehicle materiel, excluding payload and operating personnel.

6. Gross weight - weight of vehicle including fuel, lubricants, coolant, vehicle tools and spares, cargo, and operating personnel.

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DUAL WHEELS



SINGLE WHEELS

ALL DIMENSIONS A,B,C,D, REQUIRED

FIGURE 4.

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Wheels and tires:

1. Tread contact area - the area of the tread design elements in physical contact with a rigid plane surface at a given load and a specified inflation pressure.

2. Wheel tread, (see Fig. 4) - the horizontal distance at ground level between opposing tire centers of wheels on the same axle.

3. Wheelbase - the horizontal projected distance from the center of the rear axle, sprocket or idler to the center of the front axle sprocket or idler; the wheelbase of equipment with oscillating tandems is the horizontal projected distance between the center of oscillation of the tandem and the center of the front axle; the wheel base of any combination vehicle or machine is the overall projected distance from the center of the front wheels to the center of the rear wheels.

3.2 Engineering symbols. The following engineering symbols apply to the terms used in this standard:

Symbol	Term	Symbol	Term
a	Acceleration, linear	D or d	Diameter
A	Area	d.c.	Direct current
a.c.	Alternating current	deg	Degree
amp	Ampere	dm	Decimeter
avg	Average		
		E_k	Energy, kinetic
b	Breadth or width	E_m	Efficiency, mechanical
bhp	Brake horsepower	E_p	Energy, potential
bmep	Brake mean effective pressure	E_w	Work
bp	Boiling point	F	Force, concentrated
Btu	British thermal unit	F_c	Force, tractive, corrected or Drawbar pull, corrected or Draft load, corrected
°C	Temperature, Centigrade		
cc	Cubic centimeter		
cfhr	Cubic feet per hour	F_o	Force, tractive, observed or Drawbar pull, observed or Draft load, observed
cfm	Cubic feet per minute		
cfs	Cubic feet per second		
cg	Center of gravity		
CP	Center of pressure	f_c	Force, vehicle drag, corrected
cuft	Cubic feet		

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Symbol	Term
fo	Drag, vehicle or vehicle drag, observed
fhp	Friction horsepower
fpm	Feet per minute
fps	Feet per second
ft	Foot or feet
ft-lb	Foot-pound
°F	Temperature, Fahrenheit
g	Acceleration of gravity
G	Gear ratio
gal	Gallons
gph	Gallons per hour
gpm	Gallons per minute
gps	Gallons per second
h	Height or depth
hg	Mercury
hp	Horsepower
hph	Horsepower hour
hr	Hour
I	Ampere
ID	Inside diameter
in	Inch
kc	Kilocycle
kg	Kilogram
km	Kilometer
kv	Kilovolt
kva	Kilovolt-ampere
kw	Kilowatt
L	Length
lb	Pound
lb/bhp/hr	Pound per brake horsepower hour
m	Mass or meter
max	Maximum
mep	Mean effective pressure
mi	Mile
min	Minute
mm	Millimeter
mp	Melting point
mph	Miles per hour
mphps	Miles per hour per second

Symbol	Term
n	Efficiency
N	Revolutions, per unit time
n_e	Revolutions, engine, rpm
N_e	Revolutions, engine
n_p	Revolutions, propeller shaft, rpm
N_p	Revolutions, propeller shaft
n_w	Revolutions, wheel, rpm
N_w	Revolutions, wheel, rpm
OD	Outside diameter
oz	Ounce
p	Pressure, normal force/ unit area
P	Power
pfc	Pound per cubic foot
psf	Pound per square foot
psi	Pound per square inch
psig	Pound per square inch, gage
Q	Quantity in general (matter)
qt	Quart
r or R	Radius
rpm	Revolutions per minute or speed, engine
s	Distance, linear
S	Slip wheel or track, percent
sec	Second (time)
sp gr	Specific gravity
sq	Square
s_s	Distance, 1 rev of wheel, towed
s_t	Distance, total traveled
std	Standard
s_u	Distance, driving
t	Time, or time elapsed, time of run
T	Torque
THPc	Horsepower, tractive, corrected
THPo	Horsepower, tractive, reserved

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Symbol	Term
v	Volt
V	Velocity, linear or Road speed, vehicle
Vol	Volume
Vw	Wind velocity, mph
w	Watt
W	Weight or load total
yd	Yard

3.3 Hydraulic symbols:

Term	Symbol
Reservoir	
Pump	
Hand pump	
Manual valve	
Pressure gage	
Filter	
Bypass indicator	
Flowmeter	
Temperature measuring device	
Accumulator	
Motor	

Note: Some symbols may have several instruments that will serve the purpose, ie., a pressure gage or a pressure transducer may be used to measure pressure.

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4. GENERAL TEST CONDITIONS

4.1 These general test conditions apply to all tests included in Part II of this standard, except where specified otherwise in the individual tests.

4.2 Fuels, lubricants, and coolants. Standard military, fuels, lubricants, and coolants shall be used in test equipment. Suppliers shall provide a list of commercial types normally used and their corresponding military grades and classes.

4.3 Operating temperatures. The equipment shall be operated until the temperature of all components, lubricants, and coolants become stabilized before the recording of any test data.

4.4 Preparation of test item. Equipment being tested shall be prepared to insure that weight distribution, including ballast, comply with specifications and that all components function properly. Lubrication and servicing shall follow the supplier's maintenance guides. New equipment shall be properly "broken-in" under load and operating conditions recommended by the supplier so that it may be operated to full capacity without damage. Faulty adjustments or malfunctions that develop during the break-in period shall be corrected before proceeding with the test. Corrective action, however, shall not include redesign and structural changes.

5. TEST COURSES

5.1 The test courses described herein are for self-propelled and towed industrial, construction and materials handling equipment.

5.2 Level tractive ability test course. This course shall be smooth and uniform in texture with not more than 1 percent average grade. The crown slope shall not exceed 1.5 percent, measured from center to outer edge. The width of the straight portions shall be wider than the vehicle and long enough to meet the requirements of the test method.

5.3 Test course surfaces.

5.3.1 Paved surface. This is a concrete or bituminous road surface constructed on a prepared base. It shall be reasonably free from indentations, pits, depressions, and shall be adequate to support test loads without visible fracture, or displacement. Expansion joints in concrete surfaces are permitted if they comply with construction standards in width, depth, and length of spacing. All loose material should be swept off the surface prior to use.

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5.3.2 Improved surface. An improved or stabilized soil surface is constructed of soil blended with grading materials, such as sand, gravel, and crushed rock, or with Portland cement, bituminous or resinous compounds, or calcium chloride. The type of construction and blending materials shall conform to the requirements of the test or specification. The foundation or base shall be well drained and graded; surface indentations or depressions shall be not more than 3 inches deep. Permanent vertical penetration of the tread pattern of rubber tired test vehicles shall not exceed 1/4 inch; grouser penetration of tracked vehicles should be approximately 1/2 inch and in no case exceed 50 percent of the grouser depth. The surface shall be regraded, compacted, or sprinkled as necessary, to maintain required conditions (see Note 1).

5.3.3 Graded surface. This is natural soil, graded, drained, smoothed down, and compacted. It shall have no depressions more than 3 inches deep. Permanent vertical track penetration shall be not more than the depth of the grouser. The surface shall be regraded, compacted, or sprinkled, as necessary to maintain specified conditions (see Note 2).

5.3.4 Scarified surface. A scarified surface is constructed by scarifing natural undisturbed soil (see 5.3.5) with a rotary tiller, disc-harrow, or other similar equipment to a consistency whereby it may be hand-scooped to a depth of 6 inches without the assistance of handtools. This surface shall be free of excessive ruts or deep depressions so that the contour is similar to that of a graded surface.

5.3.5 Natural undisturbed surface. This is undisturbed ground, barren or covered with vegetation. Surface conditions shall be as indicated in the test or specification.

5.3.6 Surface finishes. In the test of self-propelled vehicles, test courses with the following surface finishes are commonly used:

(a) Paved
  (1) Concrete, smooth (floor finish).
  (2) Concrete, harsh (belt or broom finish).
  (3) Concrete, exposed aggregate.
  (4) Bituminous, smooth.
  (5) Bituminous, harsh (surfaced with open-type mixes).
  (6) Bituminous, chip stone surface.

(b) Improved and stabilized surfaces
  (1) Silty gravel - (over half of coarse fraction larger than No. 4 sieve size).

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(2) Clayey gravels - gravel-sand-clay mixture (over half of coarse fraction larger than No. 4 sieve size).
(3) Silty sands - sand-silt mixtures (fines with low plasticity; over half of coarse fraction smaller than No. 4 sieve size). (See Note 3.)
(4) Clayey sands - sand-clay mixtures (plastic fines; over half of coarse fraction smaller than No. 4 sieve size).

(c) Graded or natural undisturbed surfaces.

The classification of graded or natural undisturbed surfaces, shall be specified by typical name and group symbol, found in TM5-330, Planning and Design of Roads, Airbases, and Heliports in the Theater of Operations.

Note 1. The California Bearing Ratio (CBR), if used to measure surface-bearing characteristics, should be within the limits of 15 and 25 percent (see ASTM D1883).

Note 2. The CBR should be within the limits of 8 and 15 percent (see Note 1, and 5.3.2).

Note 3. Crushed rock and fines may replace gravel and sand in the same mixtures. Additives, such as Portland cement, bituminous materials (cut-back asphalt and road oils); resinous compounds, and calcium chloride or other dust palliatives, may be used to improve stabilized surfaces.

6. TEST APPARATUS

6.1 Scope. This section establishes the general standards of accuracy for suggested instrumentation prescribed in the tests. Instruments used shall be of the accuracy specified herein except in those cases where the test specifically states the accuracy of the instrumentation required. Instrumentation meeting the required accuracy may be substituted for the suggested instrumentation in this section.

6.2 Application. Test apparatus required by this standard are grouped in accordance with the phenomena to be measured as follows:

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Section	Application or Type of Measurement
6.3	Acceleration
6.4	Fluid volume
6.5	Force
6.6	Gradient
6.7	Linear dimensions
6.8	Particle counting
6.9	Pressure
6.10	Specific gravity
6.11	Speed
6.12	Stress
6.13	Temperature
6.14	Time
6.15	Weight
6.16	Wind direction and velocity

6.3 Acceleration. Acceleration or sudden change in motion, as referred to in these tests is the change in velocity per unit time: Generally measured in feet per second, each second, or as a value of the acceleration of gravity. Measurement of these effects or rates are made by use of accelerometers, with electronic amplifiers, and recorders.

Horizontal component of acceleration - the measure of speed or velocity per unit time, each second, i.e.: Number of feet of motion per second, each second. The value for rectilinear motion when translated to curvilinear motion as in "crane-boom swing" is measured in radians per sec^2 and may be expressed as either acceleration or deceleration (starting or stopping).

Vertical component of acceleration - the measure of change in velocity or speed per unit time induced by or resulting from shock, vibration, or impact. This measure is commonly expressed in terms of "g-forces", i.e. in multiples of the product of the mass of the object and the acceleration due to unbalance forces. Unit g-force = mass (m) of object X acceleration due to unbalance forces. This can be expressed as $F/W = \alpha/g$; F = force acting upon a body of weight (W); α is acceleration produced by unbalanced force (F); and g is acceleration due to gravity. (Hence an acceleration component of 10 g's = 10 W).

Type of measurement	Instrument	Accuracy	Remarks
a. Linear Acceleration	Accelerometer with: (1) electronic amplifier. (2) recorder.	± 2 percent of measured value.	

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6.4 Fluid volume. For many tests of hydraulic systems, fluid volume measurement is important. Instruments usually indicate total flow, flow per unit time, or give a dimensionless number which must be translated on a chart to give results.

Type of measurement	Instrument	Accuracy	Remarks
a. Volume (Fluid)	(1) Positive displacement flowmeter	± 1 percent of observed flow	
	(2) Turbine flowmeter	± 2 percent	
	(3) Rotometer	± 2 percent	Dependent on viscosity

6.5 Force. The act or cause which tends to or imparts motion on matter.

Type of measurement	Instrument	Accuracy	Remarks
a. Tractive effort:	(1) Chatillon gage (mechanical)	± 2 percent of observed force.	Measures force by deflection, transmitted by gear assembly to indicator with calibrated scale (usually a static or onetime measurement).
	(2) Hydraulic cylinder w/indicator.	± 1 percent of observed force.	Consisting of steel circular bar with full strain-gage wheatstone bridge installed in load cell; the applied force changes electrical resistance in the bridge and records same by means of deflection of a pen on a strip chart.
	(3) Load cell, electronic w/amplifier and recorder.	± 2 percent of observed force as recorded.	
	(4) Load cell, w/hydraulic pressure graph recorder.	± 2 percent of observed force.	Remote recording possible by use of hydraulic line.

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6.6 Gradient. The slope of an inclined plane or test course derived by dividing the height or vertical rise by the horizontal distance or length.

Type of measurement	Instrument	Accuracy	Remarks
a. Angles of inclined planes: (or % grade or slope).	(1) Inclinometer.	± 30 min. or 1/2 of a degree of arc.	Convert to % slope.
	(2) Level, surveyor's Spirit Level.	± 1 percent of measured distance. Normally greater necessary for measurements up to 100 feet.	Requires calculation, $\frac{\text{height} \times 100}{\text{distance}} = \%$ slope Note. Unlike precise surveying, the determination in these tests may be obtained by measurement to the nearest tenth of a foot for an accuracy of one-tenth of one percent.

6.7 Linear dimensions. The horizontal or distance measurements incident to the determination of length of test runs, physical dimensions of equipment, contour surfaces, etc. Cloth tapes are not satisfactory unless reinforced with several strands of steel wire (metallic tapes). If tests require coordination of distance, time, and force, an odometer or similar instrument shall be used.

Type of measurement	Instrument	Accuracy	Remarks
a. Distance measurement:	(1) Odometer	± 0.1 percent of measured distance.	Shall have wheel of known circumference with either an electrical or mechanical revolution counter, converting to direct reading.

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6.7 Linear dimensions (Cont'd).

Type of measurement	Instrument	Accuracy	Remarks
a. Distance measurement:	(2) Tape, measuring surveyors, chain.	± 0.1 percent of measured distance or 1/32 in. whichever is greater.	Should be selected and used with care, especially when this high accuracy is required.
	(3) Transit, surveyor's and rod stadia.	± 0.1 percent of measured distance.	Equipped with stadia cross hairs. Transit provides rapid measurement of test course or travel distance.
b. Horizontal and vertical dimensions:	(1) Level, carpenters.	± 1/4" in 25 feet.	Carpenters line level preferable to bos level used by masons.
	(2) Manometer.	± 1 mm.	If a high grade instrument is not available a simple vertical U-tube type unit is satisfactory, as conversions are not required.
	(3) Plumb bob.	± .005 feet.	
	(4) Transit, surveyor's and rod level.	30 sec. of an arc ± .01 foot.	Normally greater than necessary for measurements up to 100 feet.

6.8 Particle counting. The particle counter for particle counting shall be calibrated in accordance with ANSI B93.28. Channel settings shall be within 5 percent of the known sample.

6.9 Pressure. The force of gas, solid or liquid, exerted on a body so as to alter its shape or changes its volume. Four types of pressure measurements are required:

(a) Ambient air - external force of air surrounding a body in an environment.
(b) Tire pressure - internal force contained within an inflated tire.
(c) Hydraulic pressure - internal force contained within hydraulic system.
(d) Brake pressure - internal force contained within braking system.

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Type of measurement	Instrument	Accuracy	Remarks
a. Ambient air:	Barometer.	.01 inch or 1 mm.	Mercury, tube with vernier.
b. Hydraulic pressure:	Gage, pressure dial indicating (Bourdon).	± 1 percent of measured value.	
c. Tire pressure:	Gage, tire pressure.	± 3 lb in 100 lb.	
d. Brake pressure:	(Same as "b").	(Same as "b").	

6.10 Specific gravity. The density of materials (liquids) as compared or related to water at 39° F. As it pertains to these tests, and in order to establish a common base or standard, it is often necessary to express the specific gravity of fuel, lubricants, or antifreeze, etc., under the conditions specified for the test (climatological) as a ratio to water of 1.00.

Type of measurement	Instrument	Accuracy	Remarks
a. Fuel density:	(1) Hydrometer graduated scale.	± .5 percent or nearest 0.01 (ratio) i.e. 1.00 (water).	Standard readings should normally be made at 60° F, otherwise a correction should be applied. Certain types of scales require conversion for interpretation according to accepted standards (i.e. compared to water at 39° F).

6.11 Speed. The displacement or rotation of an object by revolutions per minute. As it pertains to these tests, speed is the relationship of time and distance, expressed as revolutions or distance per unit time.

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Type of measurement	Instrument	Accuracy	Remarks
a. Engine, shaft or wheel speed:	(1) Oscillator frequency. (2) Recorder, totalizing. (3) Tachometer.	± 5 percent of indicated value. ± 1 percent of measured value. ± 2 percent of indicated value.	
b. Ground-speed:	Ground wheel or speedometer.	± 2 percent of rate observed or measured.	

6.12 Stress. The force inside a body resisting change of size or shape. As it pertains to these tests, when structural parts of equipment are subjected to straining forces, the internal resistance of stress, up to the elastic limit of the material, is determined by measurement of strain on its surface and expressed as unit stress in pounds per square inch.

Type of measurement	Instrument	Accuracy	Remarks
a. Dynamic strain:	Transducer amplifier and recorder system.	± 2 percent of measured value.	Measure of stress by these instruments requires careful and skillful operation to obtain acceptable accuracy.
b. Rotational strain:	Torque pickup, and recorder.	± 2.5 percent of observed torque.	
c. Static strain:	Strain gage and analyzer.	± 1 percent of measured value.	

6.13 Temperature. Temperature measurements made in a test usually range from 0°F to 500°F. In special environmental tests, however, they may be as low as -80°F and as high as 1500°F. Accordingly, instrumentation shall be selected on the basis of (1) temperature range, (2) type; direct-reading, recording, and nonrecording, (3) accuracy, and (4) environmental conditions.

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Type of measurement	Instrument	Accuracy	Remarks
a. Temperature:	(1) Indicator, test, dial.	± 1 percent.	
	(2) Recorder.	± 1 percent.	
	(3) Temperature elements. Bourdon tube.		
	(a) Gas filled	± 1°F	Rugged, but with slow response.
	(b) Liquid filled	± 1°F	Subject to error from temperature changes along capillary tubes; accuracy not affected by fluctuations in barometric pressure.
	Alcohol		
	Mercury	± 1°F	
	(c) Vapor pressure	± 1°F	Accuracy not affected by temperature changes along capillary tube; affected by fluctuations in barometric pressure.
	(4) Thermocouples		
	(a) Chromel-alumel	± 5°F or ± 0.75 percent.	Thermocouple wires must be protected at all temperatures from oxidizing and reducing atmospheres.
	(b) Copper-constantan	± 5°F or ± 1 percent.	Resistant to contamination; holds original calibration under conditions where others fail.
	(c) Iron-constantan	± 5°F or ± 1 percent.	All shock gradients; long thermocouple wires must be eliminated to avoid error.
	(5) Thermometer, recording, resistance.	± 0.3°F throughout range.	Used when accuracy is important and effects of environmental factors can be compensated for or controlled.

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6.13 Temperature (Cont'd).

Type of measurement	Instrument	Accuracy	Remarks
a. Temperature:	(6) Thermometer, self indicating bimetallic -100° to 1000°F.	± .1 percent throughout range.	The indicating lag averages approximately 0.2 sec. per degree.
	(7) Thermometer, self indicating, liquid in glass armored and unarmored.		
	(a) Mercury filled -38°F to 760°F.	± 1°F overall.	
	(b) Alcohol filled -95°F to 150°F.	± 1°F overall.	

6.14 Time. Time is a measure of phenomenom, which is used as a basis for determining other values under specified test conditions as it pertains to these tests.

Type of measurement	Instrument	Accuracy	Remarks
a. Time:	(1) Recorder, time.	± .2 percent of measured value.	Many recorders that function in relation to time may be applied by electronic means to measurement of time. Selection should be according to availability and adaptability.
	(2) Watch, stop.	± .2 sec. in 1 minute.	The 1/5 split-second type is best. Hand should not jump forward or backward in starting and stopping; should return to original position when released.

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6.15 Weight. As pertains to these tests, weight of equipment is one of the initial determinations in any test. For most test applications, the weight of the item is the sum of the vertical loads on each point of support, wheel, track, or skid. Distribution of total weight to front and rear axles is one of several factors affecting center-of-gravity measurements and tractive ability.

Type of measurement	Instrument	Accuracy	Remarks
a. Weight:	(1) Gage, pressure, direct reading (mechanical or hydraulic).	± 5 percent of observed weight.	
	(2) Load cell, electronic w/ strain analyzer.	± 1 percent of observed weight.	May be used with crane or hoist by means of hook and cables.
	(3) Scale, platform.	± .1 percent of observed weight.	Similar to drive-on "Truck" scales, may be checked conveniently with "known" weights.
	(4) Scales, wheelload, portable (pair required for axle-load).	± .5 percent of observed weight.	Acceptable when calibrated regularly against known standard.

6.16 Wind direction and velocity. As pertains to these tests, wind direction or velocity is a secondary determination. In many tests where results may be influenced and thus require two-directional trials before zero or null-point can be established. When it is determined that excessive wind is influencing the results obtained, tests are generally delayed rather than attempting to correct for windage as in ballistics.

Type of measurement	Instrument	Accuracy	Remarks
a. Direction and velocity:	(1) Indicator, wind direction, and speed, or anemometer.	± 3 mph.	
	(2) Wind vane or wind sock.	± 3 degrees in 360° of compass.	Choice often a matter of observation.

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Custodians:

Army - ME
Air Force - 84

User activity:

Navy - YD

Preparing activity:

Army - ME

Project No. MISC-0884

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MILITARY STANDARD

TESTS FOR

CONSTRUCTION, INDUSTRIAL, AND

MATERIALS HANDLING EQUIPMENT

PART II

TESTS

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INTRODUCTION

This part of the standard contains the tests as they are to be conducted. All applicable sections of Part I shall be complied with before the test procedure is followed. Failure to prepare the test equipment properly or using inadequate instrumentation shall nullify the test. In order that these tests may be utilized properly, the following documents must be available to test personnel:

(a) Applicable specification or procurement documents specifying the requirements or conditions the commodity must be tested under.
(b) Applicable supplier's manuals or publications describing maintenance and operating performance.

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CONTENTS

PART II TESTS

Section	Content	Number
1	END ITEM TESTS	
	(a) Aircraft Loading	T-101
	(b) Center of Gravity	T-102
	(c) Crane Load Stability	T-103
	(d) Fording	T-104
	(e) High Temperature Storage	T-105
	(f) Hoist Line Speed and Power	T-106
	(g) Low Temperature Storage	T-107
	(h) Parking Brake	T-108
	(i) Precipitation	T-109
	(j) Rail Impact	T-110
	(k) Rail Transportability	T-111
	(l) Reserve Tractive Ability	T-112
	(m) Service Brake Effectiveness	T-113
	(n) Slinging Provisions	T-114
	(o) Tiedown Provisions	T-115
	(p) Vehicle Drag	T-116
	(q) Vehicle Travel Speed	T-117
2	SYSTEMS TEST	
	(a) Hydraulic Systems Cleanliness	T-201
3	COMPONENT TESTS	
	(a) Accumulator	T-301
	(b) Cylinder	T-302
	(c) Directional Control Valve	T-303
	(d) Filter Assembly	T-304
	(e) Holding Valve	T-305
	(f) Hose Assembly	T-306
	(g) Pump/Motor	T-307
	(h) Relief Valve	T-308
	(i) Reservoir	T-309

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END ITEM

TESTS

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TEST T-101

AIRCRAFT LOADING

1. SCOPE

This test is to determine whether the commodity will load into the specified aircraft.

2. TEST METHOD

2.1 Test apparatus.

2.1.1 General test apparatus.

Means to measure	Accuracy
Linear dimensions	Sec. 6.7, Part I

2.1.2 Special test apparatus. Rigid mockup of the specified aircraft ramp and cargo compartment clearance. An example is given in Figures 101-1, through 101-8.

2.2 Test procedure.

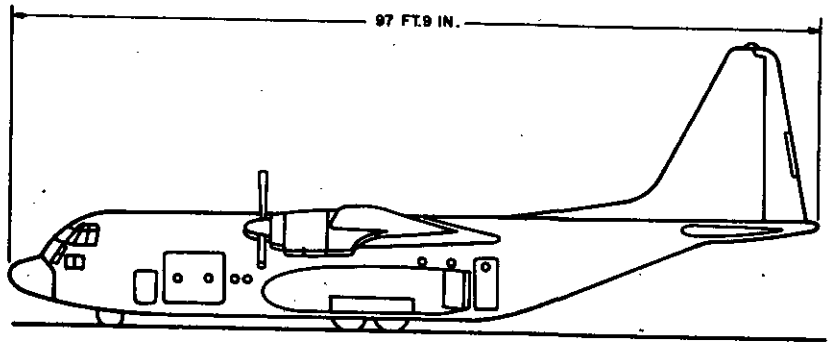
(a) Pass the commodity without attachments or as specified through the specified mockup in a direct course.
(b) Measure the vertical clearance during and after loading and the lateral clearance on each side during and after loading.

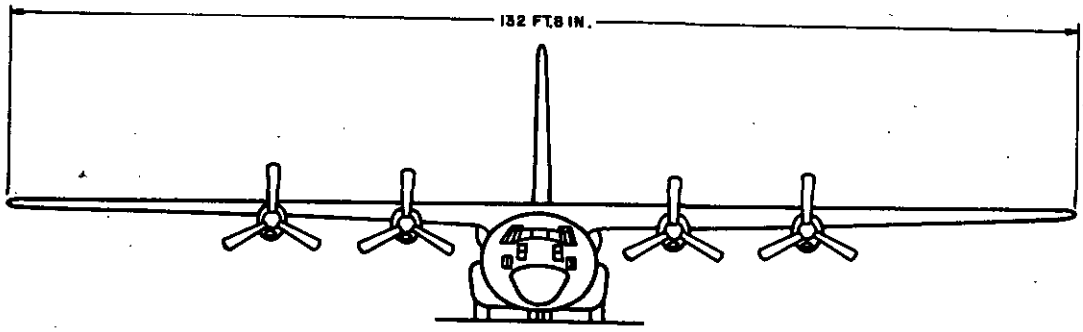
3. TEST CALCULATIONS AND RECORDS

Record the following:

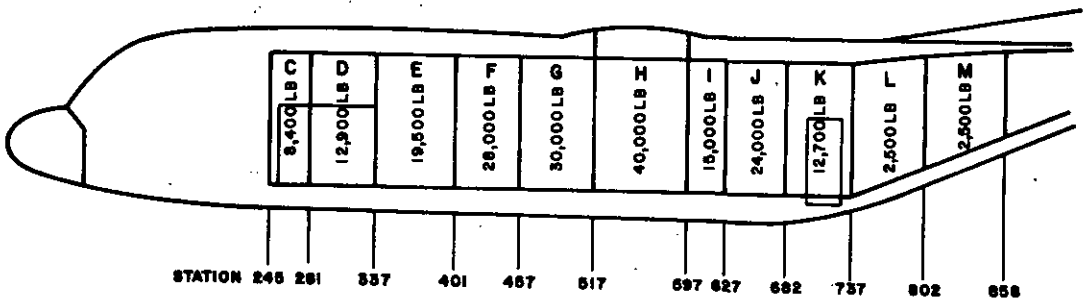
(a) Date, time, and place of test.
(b) Commodity make, model, and serial number.
(c) Clearance dimensions.
(d) Any clearance problems when the commodity runs through the mockup.
(e) Attachments and accessories removed to meet clearances.

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C-130/382B AIRCRAFT

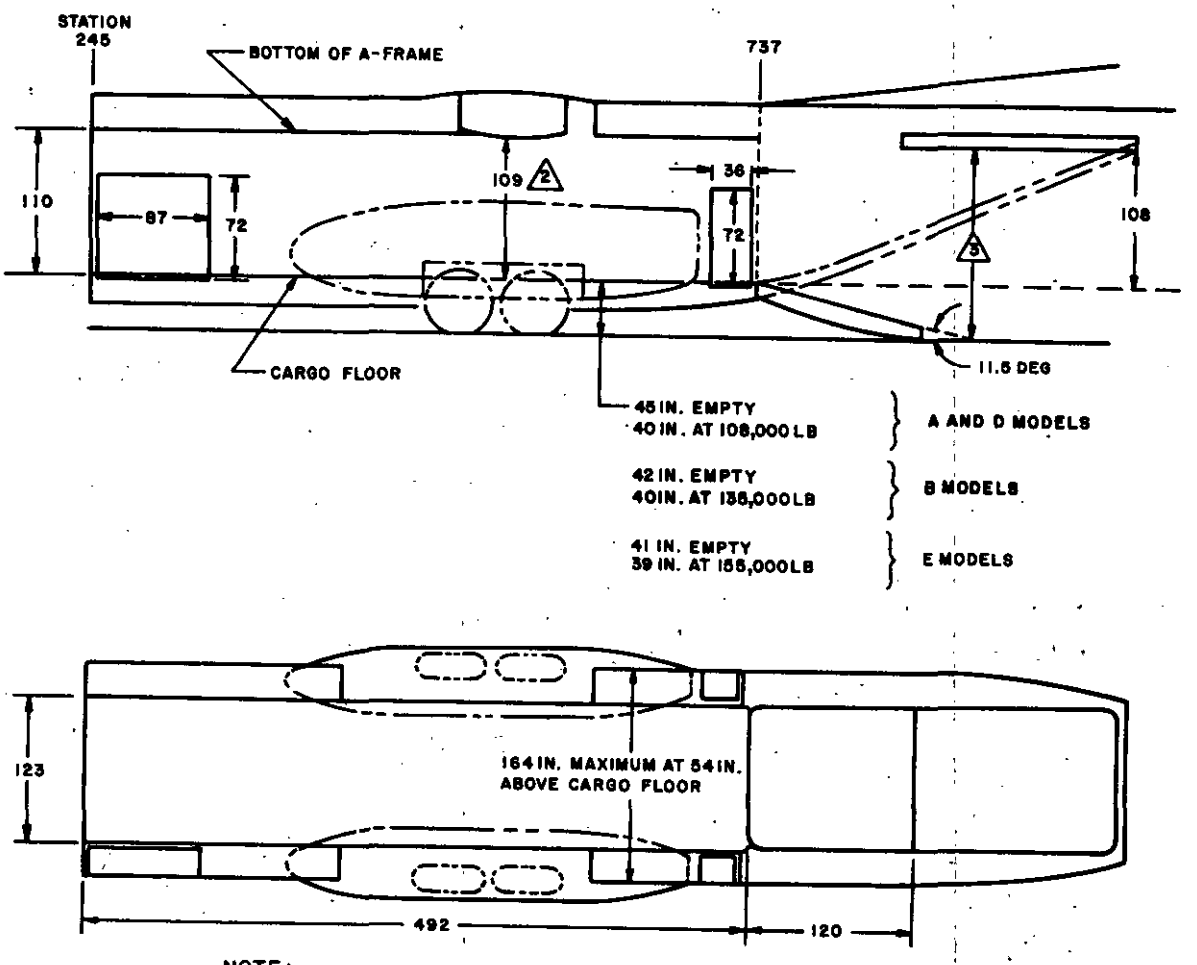


CARGO COMPARTMENT WEIGHT LIMITS
C-130/382B AIRCRAFT

FIGURE 101-1.

X-3004

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NOTE:

1. DIMENSIONS IN INCHES.
2. 108 IN. WITH LITTER BRACKET INSTALLED. WITH DUAL-RAIL AIRDROP SYSTEM INSTALLED DEDUCT 2.625 IN.
3. AIRCRAFT ON HARD SURFACE, LANDING GEAR STRUTS AT STANDARD INFLATION, NOMINAL DIAMETER TIRES AT 35 PERCENT DEFLECTION, THIS DISTANCE IS:

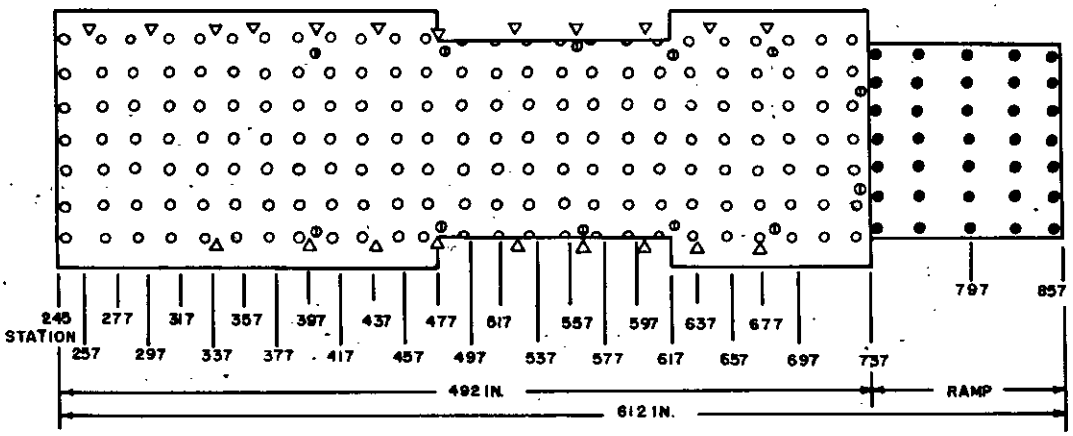
   C-130A AND D: 146 IN. AT 108,000 LB.
   C-130B: 147 IN. AT 135,000 LB.
   382B AND C-130E: 143 IN. AT 155,000 LB.

CARGO COMPARTMENT DIMENSIONS, C-130/382 B AIRCRAFT

FIGURE 101-2.

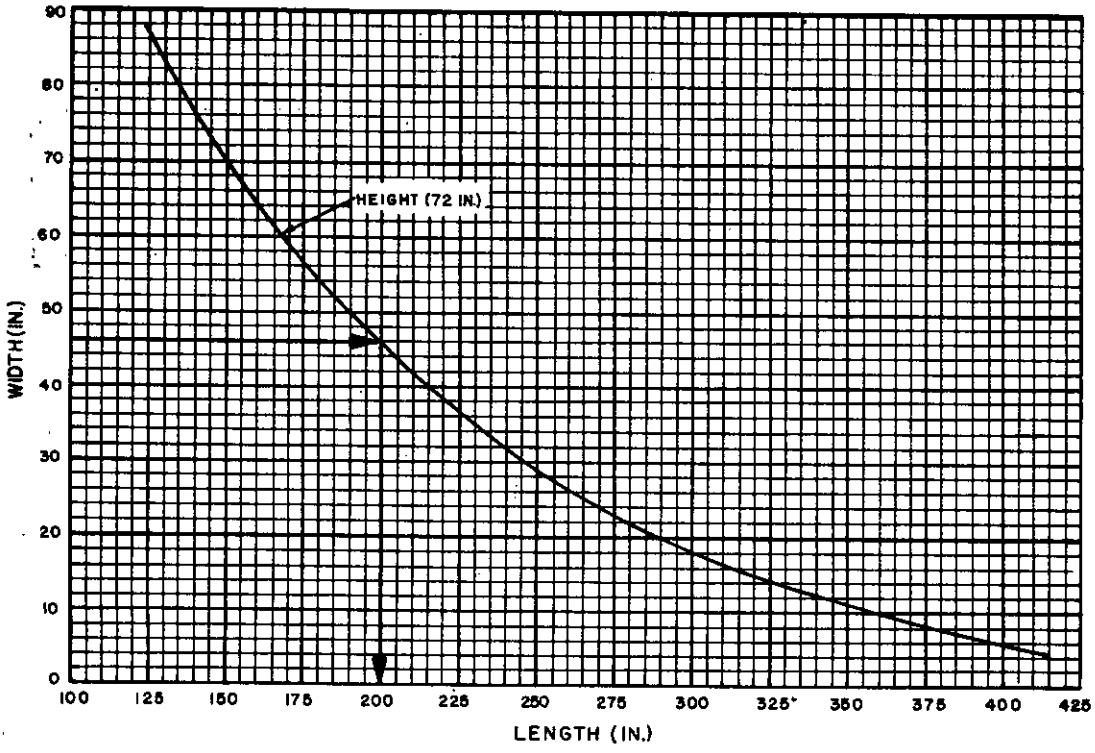
X-3005

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○ FLOOR FITTING 10,000 LB. ● RAMP FITTING 5,000 LB. ⊕ FLOOR FITTING 25,000 LB. △ SIDEWALL FITTING 5,000 LB.
⊕ RAMP FITTING 5,000 LB. 382B ONLY

ANCHORING ARRANGEMENT, C-130/382B AIRCRAFT

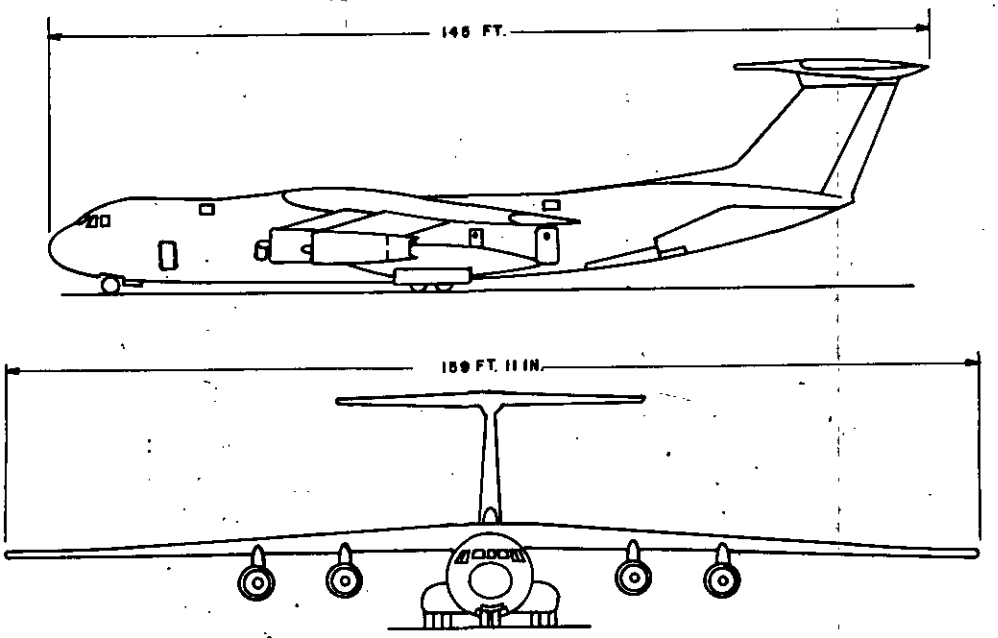


FORWARD CARGO DOOR PACKAGE SIZE GRAPH
C-130 AIRCRAFT

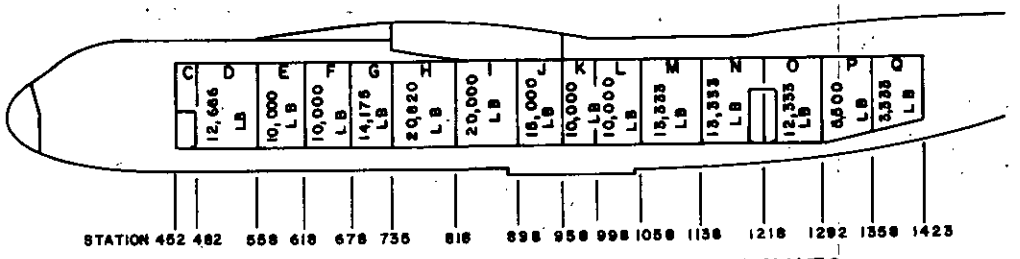
FIGURE 101-3.

X-3006

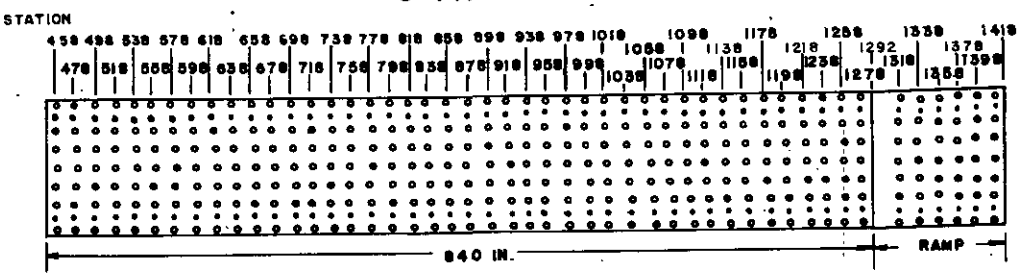
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C-141 AIRCRAFT



CARGO COMPARTMENT WEIGHT LIMITS,
C-141 AIRCRAFT



• TIEDOWN FITTING 10,000 LB.
• TIEDOWN FITTING 25,000 LB.

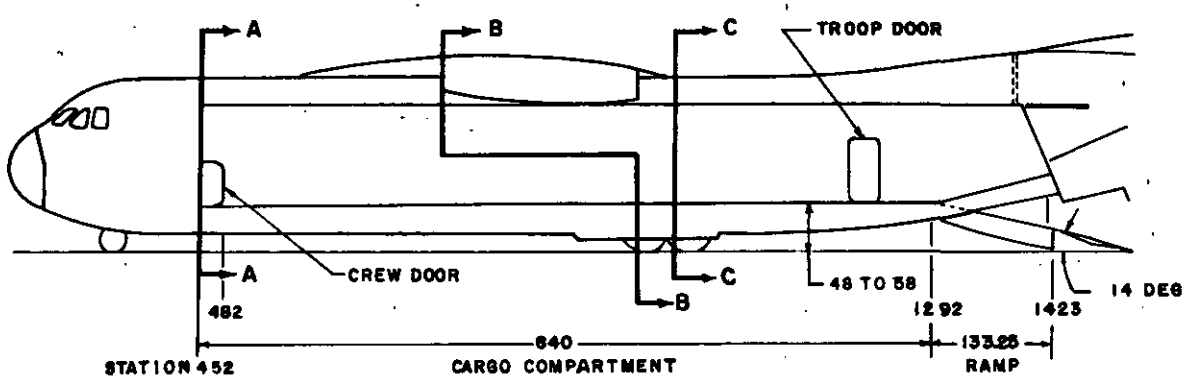
ANCHORING ARRANGEMENT
C-141 AIRCRAFT

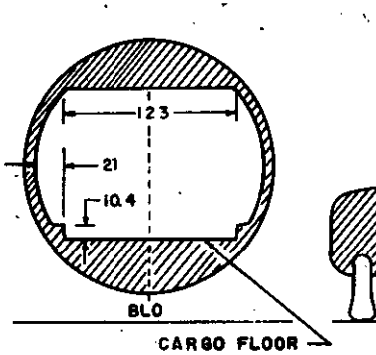
FIGURE 101-4.

X-3007

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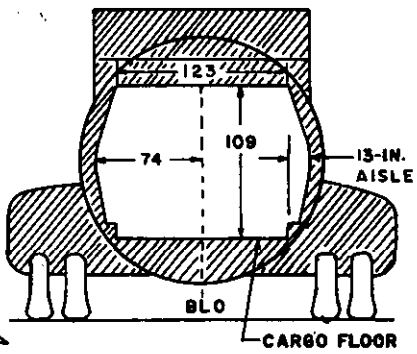
NOTES:
1. DIMENSIONS IN INCHES.
2. THE HEIGHT AND WIDTH OF THE COMPARTMENT ARE REDUCED 2 IN. ACROSS THE UPPER OUTSIDE CORNERS OF THE COMPARTMENT AT FUS STA 1196 AND 1238 BY THE TROOP DOOR TRACKS.
3. WHEN THE ROLLER CONVEYORS ARE TURNED UPRIGHT, USABLE HEIGHT WILL BE REDUCED 1.5 IN..





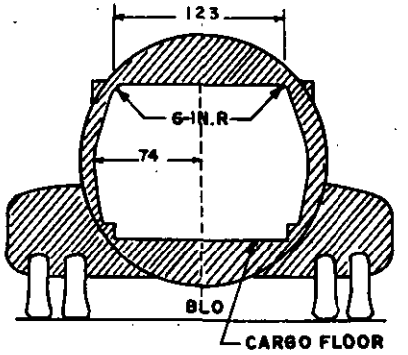
TYPICAL FROM STA 452 TO 1108 EXCEPT AS NOTED

SECTION A-A



TYPICAL AT STA 734.7 AND 958

SECTION B-B



TYPICAL AT STA 998 AND 1088

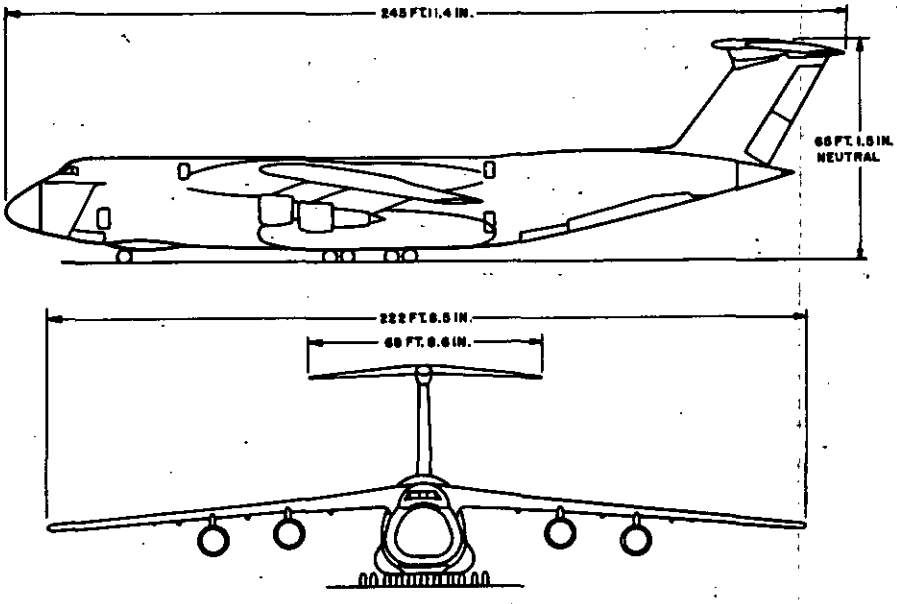
SECTION C-C

CARGO COMPARTMENT DIMENSIONS AND CONTOURS, C-141 AIRCRAFT

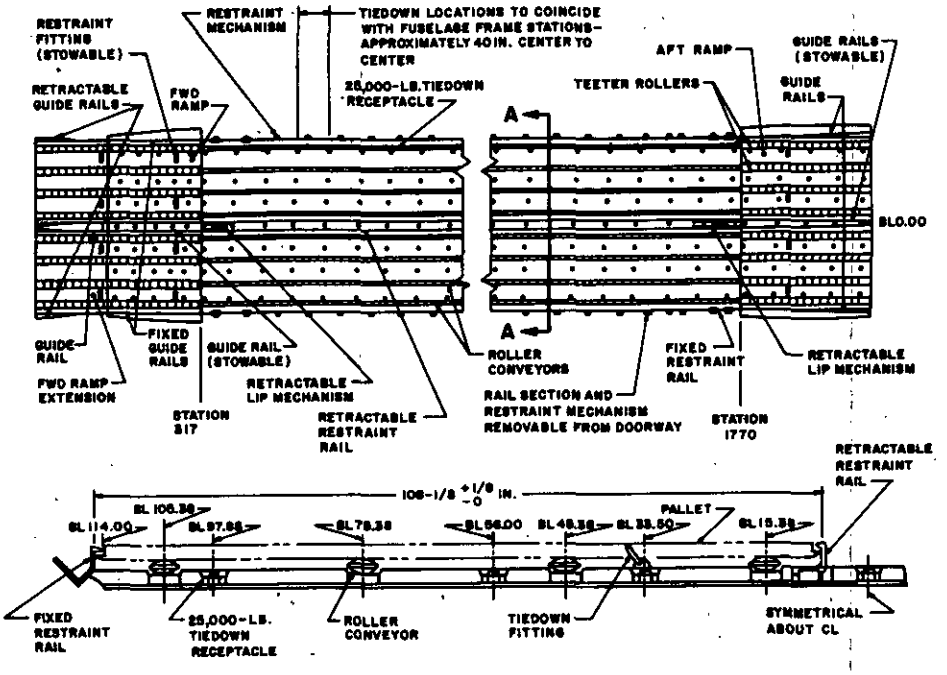
FIGURE 101-5.

X-3008

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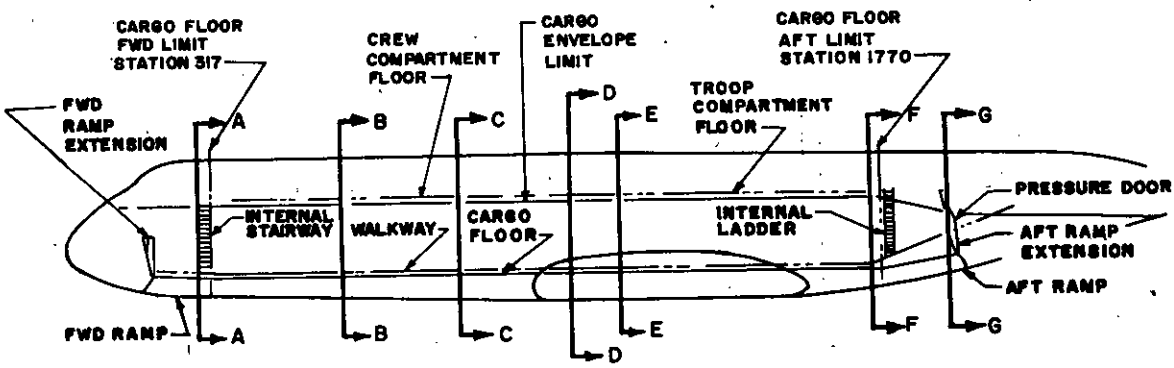
C-5A AIRCRAFT

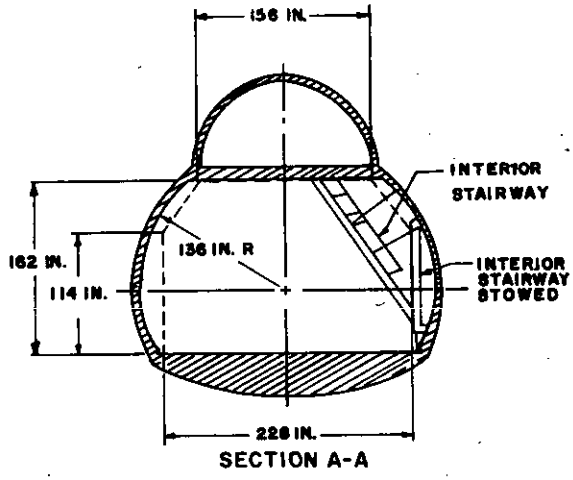


SECTION A-A ROTATED 90 DEG CLOCKWISE
ANCHORING ARRANGEMENTS, C-5A AIRCRAFT

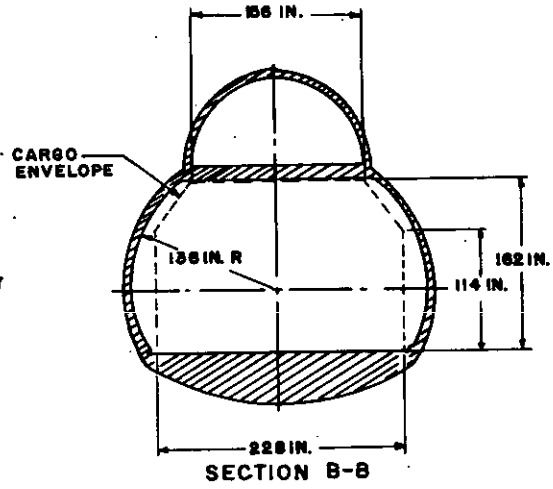
FIGURE 101-6.

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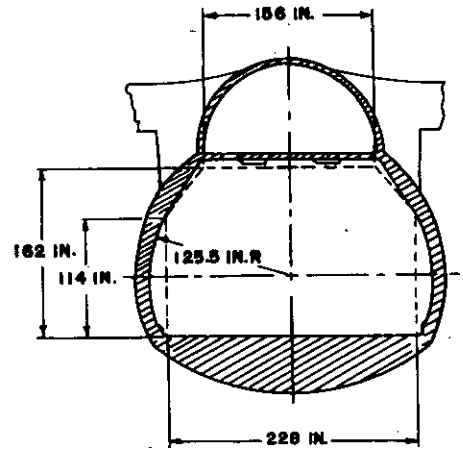




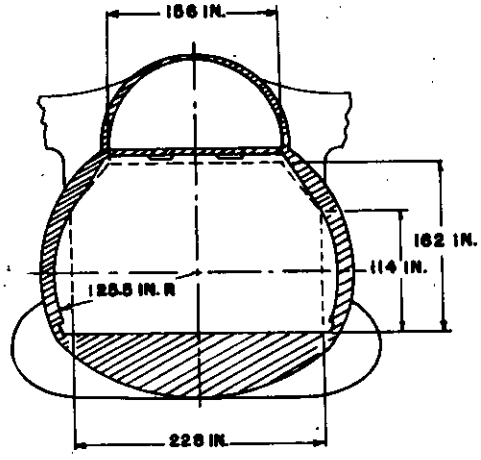
SECTION A-A



SECTION B-B



SECTION C-C



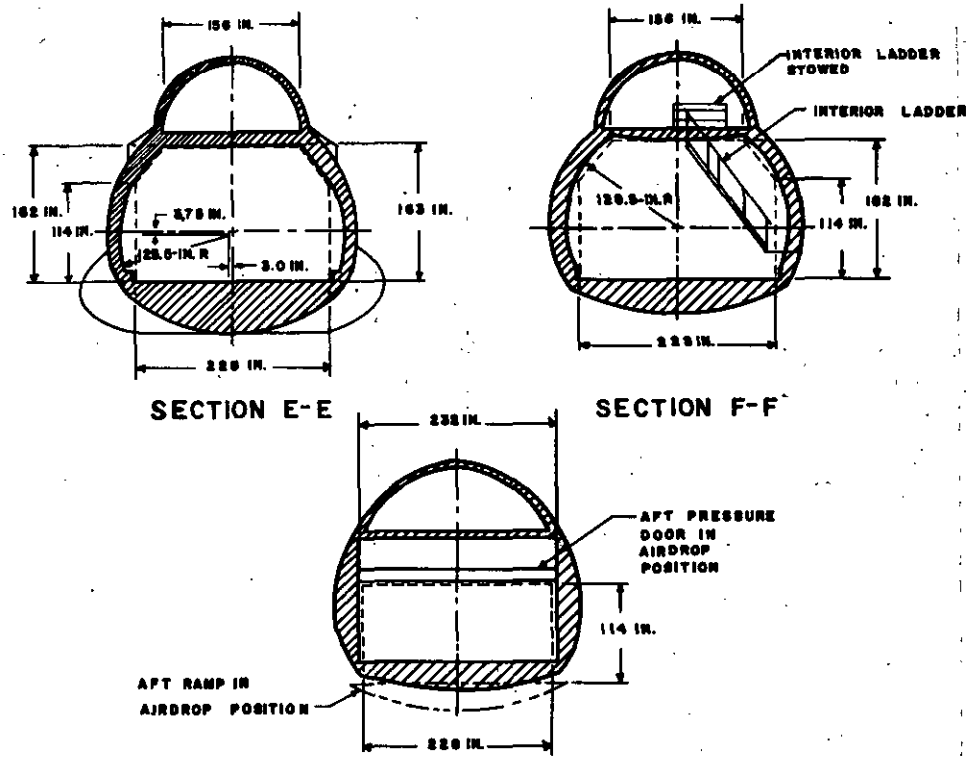
SECTION D-D

CARGO COMPARTMENT, C-5A AIRCRAFT

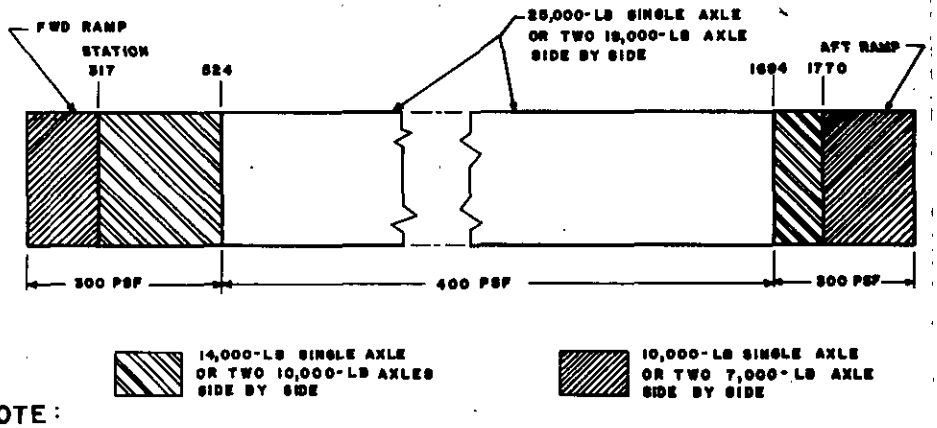
FIGURE 101-7.

X-3010

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CARGO COMPARTMENT, C-5A AIRCRAFT



NOTE:

A 25,000-LB SINGLE AXLE MAY BE DRIVEN OVER ENTIRE FLOOR AND RAMP AREA DURING LOADING OPERATIONS.

CARGO COMPARTMENT WEIGHT LIMITS
C-5A AIRCRAFT

FIGURE 101-8.

X-3011

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TEST T-102

CENTER OF GRAVITY

1. SCOPE

This test establishes the location of the center of gravity of commodities. Its application is to determine the following:

(a) Dynamic behavior of commodities.
(b) Locations for lifting or tiedown attachments.
(c) Transportation and handling characteristics.

Measurements derived through its use are presented in three dimensions from mutually perpendicular reference planes (see Fig. 102-1).

2. TEST METHOD

2.1 Limitations. Service conditions for different types of equipment may vary and thus influence final measurements because of tire inflation, location of operating attachments or other components, weight, and distribution of ballast or payload, and quantity and type of fuel, coolant, and lubricant.

2.2 Test apparatus.

2.2.1 General apparatus.

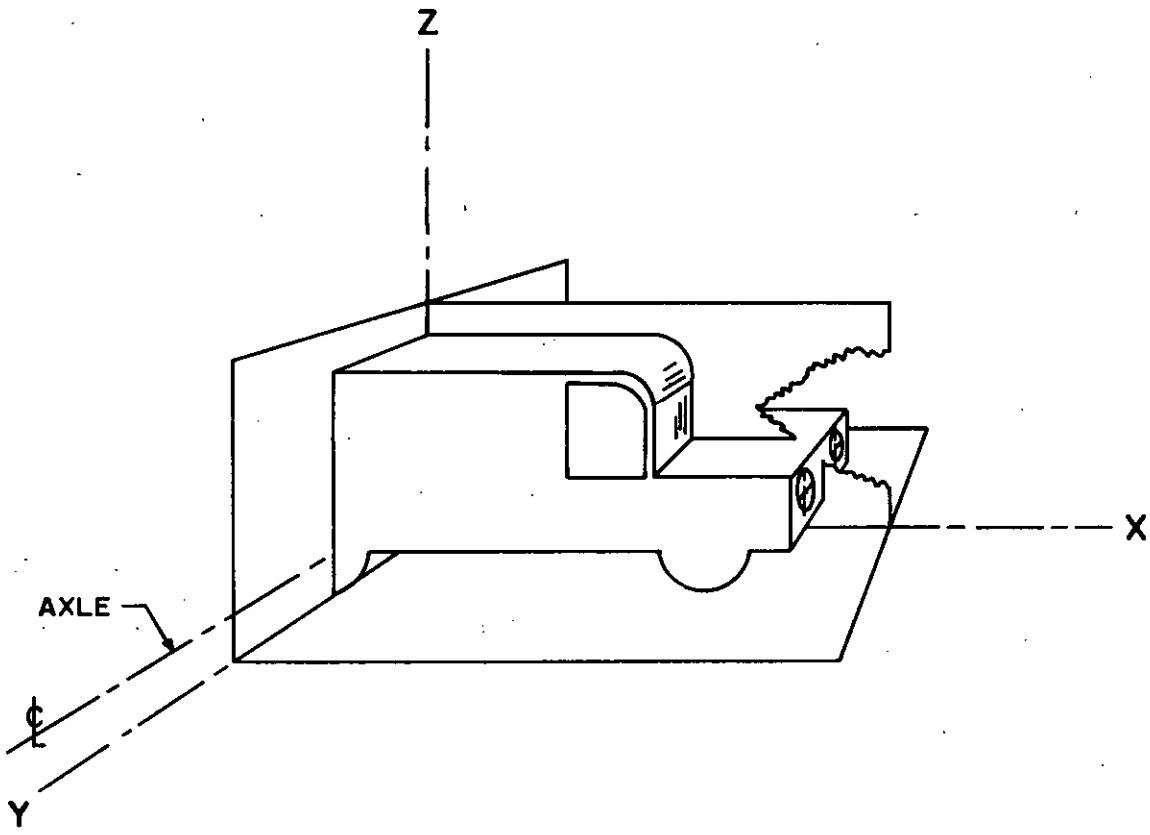
Means to measure	Accuracy
(a) Level of transverse axis	Sec. 6.7 b
(b) Projecting lines from inaccessible planes	Sec. 6.7 a
(c) Linear dimensions	Sec. 6.7 a
(d) Projecting lines or scribing board	Sec. 6.7 a

2.2.2 Special apparatus.

Instrument

(a) Crane or other lifting device.
(b) Adjustable or various lengths or slings, hook-block, and other rigging.
(c) Means to block, bolt, or chain movable components securely.
(d) Scribing board or sheet plywood faced with paper. Should be not less than 24 inches square.

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A. XY-PLANE; HORIZONTAL PLANE ON WHICH THE VEHICLE RESTS WITHOUT PENETRATION (SEE TEST COURSES-PART I, SEC. 5, PAR. 5.2).
B. XZ-PLANE; VERTICAL PLANE THROUGH THE CENTERLINE OF THE GAUGE OF THE COMMODITY TREADS.
C. YZ-PLANE; VERTICAL PLANE THROUGH AN AXLE CENTERLINE.

FIGURE 102-1. THE XY, XZ, AND YZ PLANES OF MEASUREMENT.

X-2744

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2.3 Preparation of equipment.

(a) Secure commodity components against shifting or changes in position. The lifting crane and hook-block, sling, and other rigging shall be carefully analyzed for symmetry or balance to insure their effect on commodity balance and assure safety in lifting operations.
(b) Clean commodity and remove extraneous material or residue dirt.
(c) Secure all accessories in their specified positions, and all liquids against spillage, in a manner which will not appreciably affect the distribution of weight and thus the test results.
(d) Attach the scribing board or sheet or plywood to side of commodity parallel with the XZ plane and center it approximately over the transverse projection of the anticipated center of gravity (see Fig. 102-2).
(e) Obtain a spirit level or manometer and prepare to check the transverse level of suspended commodity on reference line or transverse axis which has been established prior to suspension for each position of suspension.
(f) Vehicle weight shall be as specified.

2.4 Test procedure.

(a) Attach lifting apparatus and, in successive operation suspend the commodity with its transverse axis level and its longitudinal axis:

(1) Approximately level.
(2) Up at least 15 degrees toward the front, and
(3) Up at least 15 degrees toward the rear.

(b) Locate the plane of gravity, passing through the point of suspension, and perpendicular to the X2 plane. Project this plane to scribing board and mark a vertical line for each suspended position (see Fig. 102-2).
(c) Before actual scribing, make sure that the transverse axis in each case is level, and that the line of sight is perpendicular to the longitudinal axis of the commodity.
(d) The projection of the vertical planes on the scribing board will form, for the three positions, a triangle of error the geometric center of which is one of the points for establishing the center of gravity.
(e) Measure and record the distances from this geometric center of the intersection points (or intersection lines) to the XY and XZ reference plane of the commodity.

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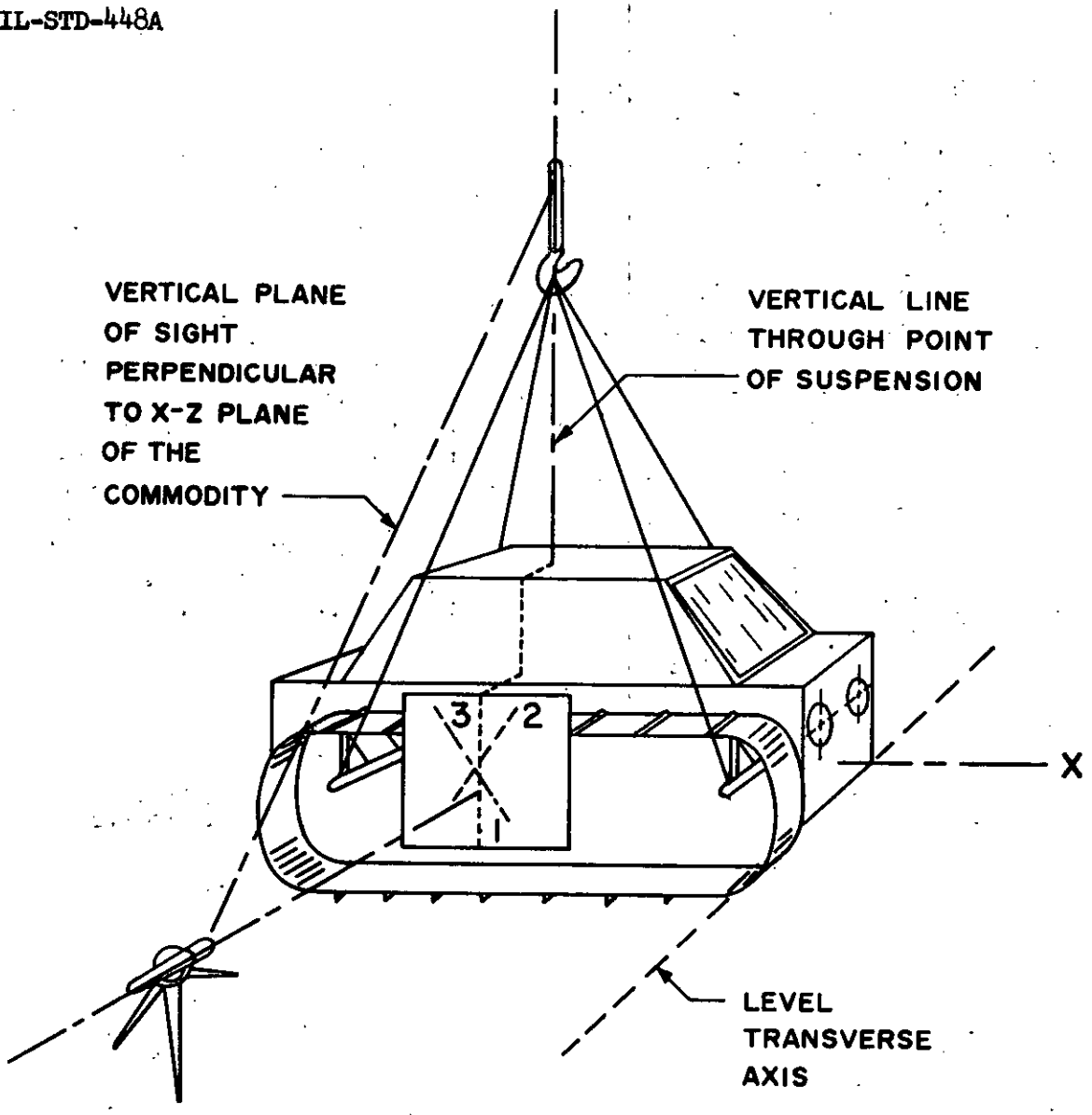


FIGURE 102-2. THE LOCATION OF THE XZ, XY, AND YZ PLANES OF MEASUREMENT BY THE SUSPENSION METHOD.

X-2745

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(f) Measure and record the distance between each point of intersection on scribing board and the actual geometric center of all intersection points. These distances are an indication of inaccuracies in technique and shall not exceed 2.4 percent of the perpendicular distance from the geometric center to the XY reference plane of the commodity.

(g) To establish the Y and Z coordinates of the center of gravity use the same procedure outlined in (a) through (g) with the following exceptions:

(1) Keep longitudinal axis of the commodity level;
(2) Tilt the transverse axis; and
(3) Center the scribing board over the longitudinal projection of the anticipated center of gravity and parallel to the YZ plane.

(h) Record the average dimensions of the Z coordinates for the two determinations on the test summary sheet. The difference between the two determinations should not exceed 2 percent of the average.

3. TEST CALCULATIONS AND RECORDS

Fill out and record the data as required on the data sheet. (see Fig. 102-3).

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CENTER OF GRAVITY TEST SUMMARY SHEET

GENERAL DATA

COMMODITY DESCRIPTION:

Make _____

Model _____ Serial No. _____

TIRES:

Front Size _____ Ply _____ Psi _____

Intermediate Size _____ Ply _____ Psi _____

Rear Size _____ Ply _____ Psi _____

FUEL _____ LUBRICANTS _____ COOLANT _____

ATTACHMENTS: Name _____

Location _____

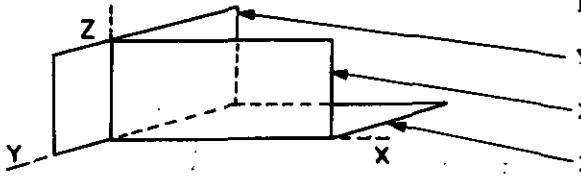
BALLAST AND PAYLOAD: Description _____

_____

Payload _____

TEST LOCATION _____

TEST METHOD USED: _____



Reference Planes Location:

YZ Plane _____

XZ Plane _____

XY Plane _____

TEST RESULTS

Coordinates of center of gravity with reference to planes:

x _____ in. y _____ in. z _____ in.

REMARKS: _____

_____ _____
DATE

_____ _____
SIGNED

FIGURE 102-3. Test Summary Sheet X-2746

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TEST T-103

CRANE LOAD STABILITY

1. SCOPE

This test determines the maximum capacity of a crane to counterbalance loads applied on its hook block. The two methods covered for conducting this test are:

(a) Suspended - load method
(b) Anchor - load method

2. TEST METHOD

2.1 Limitations. These tests shall be used only for those load ratings which are based on stability factors and are not applicable to those ratings which are based on structural competence. The testor should take care o assure that these tests are made only in the least stable direction for the rating under test.

2.2 Test apparatus.

2.2.1 General test apparatus for both suspended-load and anchor-load methods.

Means to measure	Accuracy
(a) Linear dimensions	Sec. 6.7 a
(b) Pressure	Sec. 6.9 c
(c) Weight	Sec. 6.15 a(4)

2.2.2 Special test apparatus for both suspended-load and anchor-load methods.

(a) Means for projecting the crane axis of rotation to the test course surface.
(b) Means for measuring the horizontal distance from the axis of rotation to the center of gravity of the load.

2.2.3 Additional test apparatus for suspended-load method only.

(a) Test weights, as required to make up specified loads, and to provide additional load in 10 pound increments.

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2.2.4 Additional test apparatus for anchor-load method only.

(a) Means to measure the force in pounds exerted by the hoist line on the hoist anchor; accuracy ± 1/2 percent of the force measured.
(b) Means to determine that hoisting lines are vertical.

2.3 Test course. A concrete or other firm supporting surface, level within ± 1 percent of grade.

2.4 Preparation of equipment. Service and adjust the crane to specified tolerances at the conclusion of the limbering up operation.

2.5 Test procedure.

2.5.1 General procedure.

(a) Locate the crane on the test course in position for loading and lock the travel brakes.
(b) Set outriggers, if used, and jack the crane to a position where the tires or tracks within the boundary of the outriggers are unloaded.
(c) Vertically project the superstructure axis-of-rotation to the surface of the test course and mark its location.

2.5.2 Procedure for suspended loads.

(a) Prepare test load including test weights, hook block, slings and other auxiliary equipment such as load basket, that make up the specified load weight within ± 1 percent. Record this value.
(b) With the crane superstructure in the specified position, hoist the load free of the test course at a radius where the crane is stable, then boom the load out to a radius near the balance point.

Note. The load should be kept near the test course surface in order to avoid excessive tipping of the crane. Also the crane should be safeguarded by blocking or other means from overturning backward should the load line or other tackle fail while under load.

(c) Alternately measure the radius of load and add a 10 pound increment to the load until the load overcomes the stability of the crane. The radius of load and load weight last obtained, before the load overcame the stability of the crane, shall be recorded as the balance point condition.

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2.5.3 Procedure for anchor-load method.

(a) Determine the weight of the hook block and any part of the force measuring means that will be suspended by the hook block. Record these values.
(b) Install the force measuring means between the hook and anchor.
(c) With the crane in the specified position for lift and with the load line kept vertical while under load.
(d) Apply hoisting force to the anchor until the indicated magnitude of force tends to diminish with continued hoisting.
(e) Record the observed hoisting force and the radius of load. The peak hoisting force in pounds plus the weight of the hook block and other equipment suspended between the hook block and anchor constitutes the load for record.

3. TEST CALCULATIONS AND RECORDS

3.1 Capacity curve. Where it is desired to determine the balance point capacity of a crane throughout a range of loads or radii, follow procedures as outlined for individual determinations, making sure that load and radius are determined for each extreme of the range and at a sufficient number of intermediate points to permit plotting a curve. Plot a curve showing the maximum capacity of the crane with the load in pounds as ordinate and radii in feet as abscissa.

3.2 Test records. Record a description of the crane, positions for test, load data and radius of load data on the Physical Dimensions Test Summary sheets (see Figs. 103-1, 103-2, and 103-3).

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CRANE LOAD STABILITY TEST SUMMARY

Testing Agency_____ Location_____
Make_____ Model_____ Serial NO._____
Condition: New_____ Used_____ Hours_____
Mounting: Type_____ Size_____
Tires: Size_____ Ply_____ Pressure (psi)_____
Boom: Type_____ Length (ft)_____
Boom Jib: Type_____ Length (ft)_____ Angle to the Boom_____
Engine: Make_____ Model_____ Serial No._____
Counterweight: Type_____ lb._____
Test Method: Suspended Weights_____ Anchor_____

BALANCE POINTS

Position of Superstructure	Without Outriggers		With Outriggers	
	Load (lb)	Radius (ft)	Load (lb)	Radius (ft)

Remarks

Test Engineer:_____ Date of Test_____

Figure 103-1. Page 1 of Test Summary X-2747

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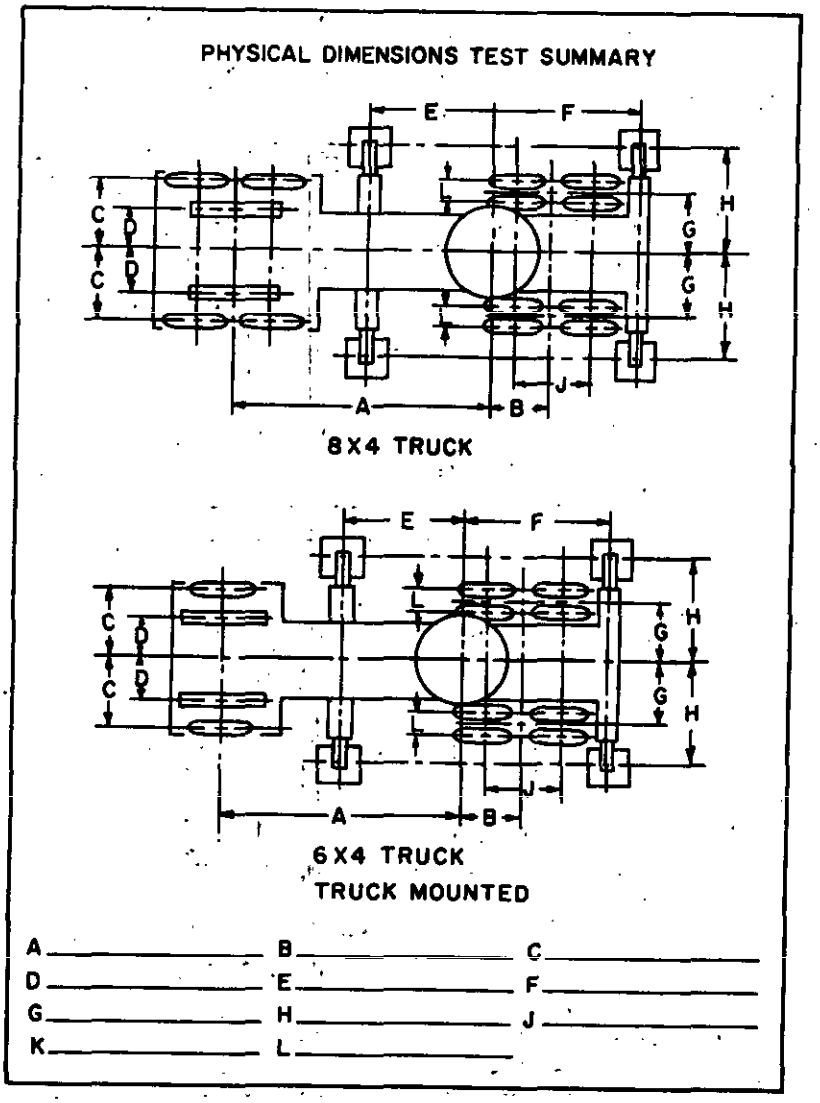


FIGURE 103-2. PAGE 2 OF TEST SUMMARY

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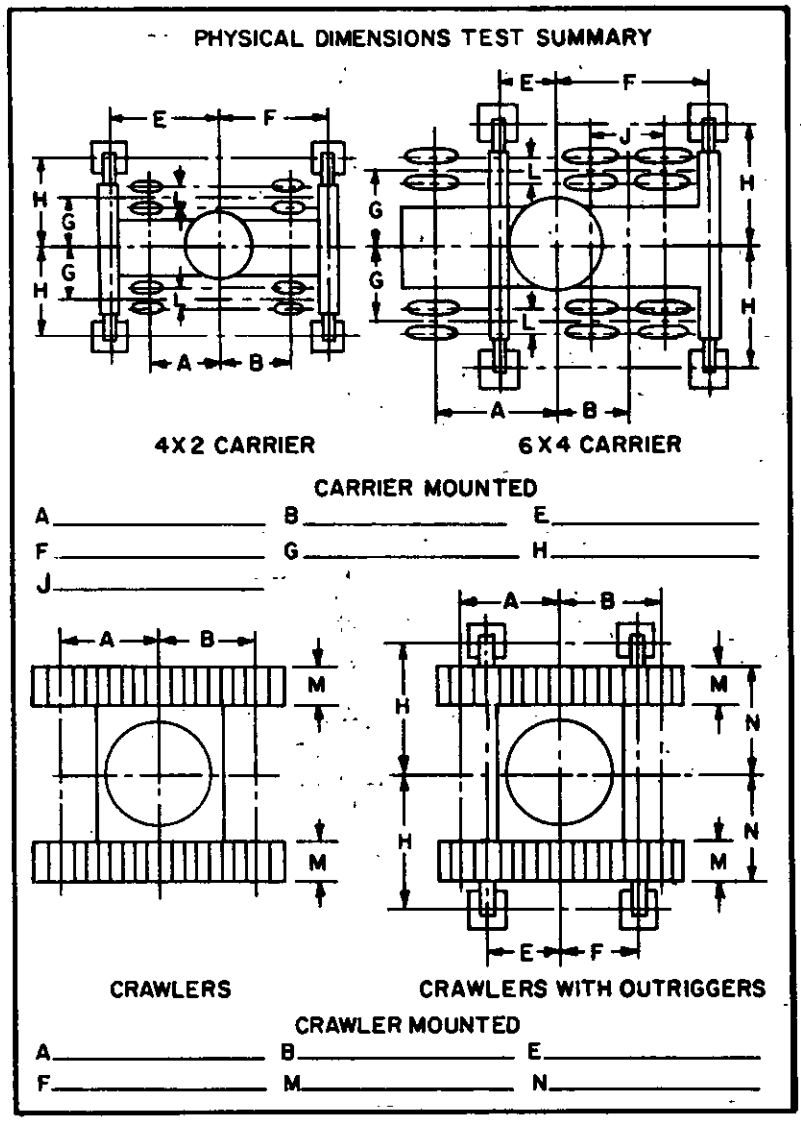


FIGURE 103-3. PAGE 3 OF TEST SUMMARY

X-2749

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TEST T-104

FORDING

1. SCOPE

This test is to determine whether the commodity can operate efficiently and safely in bodies of water.

2. TEST METHOD

2.1 Test apparatus.

2.1.1 General test apparatus.

Means to measure	Accuracy
(a) Time	Sec. 6.14 a
(b) Linear dimensions	Sec. 6.7 a

2.1.2 Special test apparatus.

(a) Oil containers.
(b) Suction pump.
(c) Instrument for analyzing water content.

2.2 Test course.

(a) Body of water not less than that specified in the commodity specification and of sufficient size to permit the commodity to operate 20 feet forward and 20 feet in reverse.

2.3 Test procedure.

(a) Prior to the test, take oil samples from the engine and all hydraulic and lubricant reservoirs for analysis as to water content.
(b) Bring the commodity engine and driveline components to operating conditions, as indicated by the temperature gages of the commodity, before the commodity enters the water. Then operate the commodity in low gear forward and reverse a minimum of 15 minutes in water at the specified depth as indicated by a line marked on the commodity.

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(c) Perform the above test a total of five times, except that during the second, third, fourth, and fifth tests, the commodity engine shall be started and stopped a minimum of two times while in the water at the specified depth.
(d) On completion of the water-fording tests, park the commodity on a dry surface, shut off the engine, and allow the commodity to stand for a period of 1 hour. Then take oil samples from the engine and all hydraulic and lubricant reservoirs.
(e) Analyze oil samples for water content.

3. TEST CALCULATIONS AND RECORDS

Record the following:

(a) Date, time, and place of test.
(b) Commodity make, model, and serial number.
(c) Record where each oil sample was taken and water content of oil before and after test.

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TEST T-105

HIGH TEMPERATURE STORAGE

1. SCOPE

This test is to determine if a commodity can be safely stored at high temperatures as specified.

2. TEST METHOD

2.1 Test apparatus.

2.1.1 General test apparatus.

Means to measure	Accuracy
Temperature	Sec. 6.13 a

2.1.2 Special test apparatus. Test chamber for specified temperatures.

2.2 Test procedures.

(a) Place the commodity in a test chamber of sufficient size and capacity to contain the commodity and to provide the specified storage temperature.
(b) Subject the commodity to a high ambient temperature until the temperature of all parts is stabilized at not lower than the specified temperature. Figure 105-1 contains environmental conditions requirements; Categories 2, 5, 6, and 7 are recommended for construction type equipment. Maintain the specified temperature such that all parts obtain it for the specified period of time.
(c) Examine the commodity.
(d) Return the commodity to ambient temperature, start the engine and operate all controls.
(e) Examine the commodity.

3. TEST CALCULATIONS AND RECORDS

Record the following:

(a) Date, time, and place of test.
(b) Commodity make, model, and serial number.
(c) Any problems or failures of the commodity due to high temperature exposure.

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Climatic Category	Operational Conditions: Ambient Air Temperature 4' -6' above ground (°F)	Operational Conditions: Solar Radiation BTU/FT^2/HR	Operational Conditions: Ambient Relative Humidity %	Storage and Transit Conditions: Induced Air Temperature Max. (°F) No solar radiation	Storage and Transit Conditions: Induced Relative Humidity %
1 Wet - Warm	Nearly constant 75	Negligible	95 - 100	80 - over 1 day	95 - 100
2 Wet - Hot	78 to 95 4 hours at 95	0 - 360	74 - 100	160 - 1 hour 155 - 4 hours negligible wind	10 - 85
3 Humid - Hot, Coastal Desert	85 to 100 Not more than 4 hours at 100	0 - 360	63 - 90	same as Category 2	10 - 85
4 Hot - Dry	90 to 125 4 hours above 120, 1 hour at 125	0 - 360	5 - 20	same as Category 2	2 - 50
5 Intermediate Hot - Dry	70 to 110 4 hours above 105 1 hour at 110	0 - 360	20 - 85	145 - 1 hour negligible wind	5 - 50
6 Intermediate Cold	-5 to -25 6 hours at -25	Negligible	Tending toward saturation	-30 - 6 hours no wind	Tending toward saturation
7 Cold	-35 to -50 6 hours at -50	Negligible	same as Category 6	-50 - 6 hours same as operational conditions	
8 Extreme Cold	-60 to -70 6 hours at -70	Negligible	same as Category 6	-70 - 6 hours same as operational conditions	

FIGURE 105-1. Operational, Storage And Transit Conditions Extract From AR 70-38

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TEST T-106

HOIST-LINE SPEED AND POWER

1. SCOPE

This test determines the hoist-line speed that a commodity, such as a crane, develops while hoisting a specified load with a specified condition of rope layers. Values derived may be used to calculate hoist line power. This test can be used to determine the hoist line speed and power of any hoist line mechanisms, if load can be held constant and hoist line travel distance is sufficient for the accuracy of the line speed measurements prescribed.

2. TEST METHOD

2.1 Limitations. Supplier's maximum load recommendations for the conditions of test shall not be exceeded. All values are reported as observed and should not be compared to other tests conducted under different conditions.

2.2 Test apparatus.

2.2.1 General test apparatus.

Means to measure	Accuracy
(a) Line travel distance	± 1%
(b) Line travel time	± 1%
(c) Load line force	Sec. 6.5 a
(d) Weight	Sec. 6.15 a
(e) Speed	Sec. 6.11 a

2.2.2 Special test apparatus. Test weight or other loading means as required to make up the specified load within ± 0.5 percent of the total weight specified.

2.3 Preparation of equipment.

(a) After the commodity is limbered-up, recheck all adjustments and bring to within supplier's recommended tolerances.

(b) Arrange the hoisting tackle on the machine in accordance with the supplier's instructions, using the minimum parts of hoist line necessary to assure safe and smooth operation while hoisting the specified load.

Note. It is desirable to use a single part line in order to reduce tackle friction; however, safety and load travel distance may dictate need for multiple part lines.

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2.4 Test procedure.

(a) Assemble a test load as specified.
(b) Attach the specified load to the hoist line and install line speed and force sensing and recording instruments as required.
(c) Perform sufficient trial runs of hoisting the specified load to assure that:
  (1) The machine operator is familiar with procedure of operation.
  (2) The machine is warmed up to normal operating temperatures.
  (3) The instruments are functioning properly.
(d) With the engine operating at full throttle smoothly engage the hoist and lift the load. While the load is being lifted, measure and record:
  (1) Hoisting force.
  (2) Hoisting line travel distance.
  (3) Time.
(e) Repeat this operation to obtain a total of five recorded runs.

3. TEST CALCULATIONS AND RECORDS

Fill in Test Data Summary Sheet (see Fig. 106-1), and calculate the following:

(a) For each test run, calculate the average hoist line speed, using only that portion of the records where the recorded hoisting force remains constant within + 2 percent of average. Record the average speed and load value for each recorded run on the test summary sheet.
(b) Calculate the hoist line power in horsepower for each recorded run as:

$$HP = \frac{LS}{33{,}000\ P}$$

Where L = Load (lb)
S = Line Speed (fpm)
P = Number of parts in the hoist line

Record these values on the test summary sheet.
(c) Calculate the average hoist line power for all recorded runs and record this value on the test summary sheet.

Note. Wherein the horsepower for any individual run deviates more than ± 2 percent from the arithmetic mean for all runs, a determination should be made of reasons for erratic performance, the data discarded, and the test rerun.

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HOIST LINE SPEED AND POWER TEST SUMMARY

PLACE OF TEST _____ DATE _____

CRANE MODEL _____ SERIAL NO. _____ MFR. _____

ENGINE MODEL _____ SERIAL NO. _____ MFR. _____

MEASURED RPM _____ RATED RPM _____

TYPE OF DRUM _____ DRUM DIA _____ CABLE DIA _____

NUMBER OF ROPE LAYERS USED FOR TEST _____

RIGGED FOR _____ HOIST LINE PARTS BOOM LENGTH _____

TEST NO.	LOAD (lb)	LINE FORCE (LB-AV)	LINE SPEED (FPM-AV)	HP	DEVIATION FROM AVERAGE
1					
2					
3					
4					
5					
AVERAGE					

REMARKS:

_____
TEST ENGINEER

FIGURE 106-1. Test Data Summary Sheet

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TEST T-107

LOW TEMPERATURE STORAGE

1. SCOPE

This test is to determine if a commodity can be safely stored at low temperatures as specified.

2. TEST METHOD

2.1 Test apparatus.

2.1.1 General test apparatus.

Means to measure	Accuracy
Temperature	Sec. 6.13 a

2.1.2 Special test apparatus. Test chamber for specified temperatures.

2.2 Test procedure.

(a) Place the commodity in a test chamber of sufficient size and capacity to contain the commodity and to provide the specified temperatures.
(b) Subject the commodity to a low ambient temperature until the temperature of all parts is stabilized at not higher than the specified temperature (see Fig. 105-1). Maintain the specified temperature for all parts continuously for the specified period of time.
(c) Examine the commodity.
(d) Return the commodity to the ambient temperature, start the engine and operate all controls.
(e) Examine the commodity.

3. TEST CALCULATIONS AND RECORDS

Record the following:

(a) Date, time, and place of test.
(b) Commodity make, model, and serial number.
(c) Any problems or failures of the commodity due to the low temperature exposure.

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TEST T-108

PARKING BRAKE

1. SCOPE

This test determines the holding force of the parking brake of a vehicle on a level surface and a specified required grade.

2. TEST METHOD

2.1 Test apparatus.

2.1.1 General test apparatus.

Means to measure		Accuracy
(a)	Tire pressure	Sec. 6.9 b
(b)	Vehicle weight	Sec. 6.15 a
(c)	Force to apply parking brake	Sec. 6.9 d
(d)	Brake holding force (line pull)	Sec. 6.5 a
(e)	Gradient	Sec. 6.6 a

2.1.2 Special test apparatus. Towing vehicle(s) with sufficient power and weight to pull the test vehicle as required.

2.2 Test course. The level test course shall be a clean swept, dry, brushed concrete surface or other specified surface with a grade not more than ± 1 percent with sufficient length to perform test adequately. The percent grade to conduct test on shall be as specified.

2.3 Test procedure.

2.3.1 Level surface.

(a) With the test vehicle engine disengaged from the drive train; apply the parking brakes.
(b) With the line of draft horizontal with the longitudinal axis of the vehicle; apply a pull first to the front and then to the rear of the vehicle until the brake wheels slide or rotate.

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2.3.2 Percent grade.

(a) Drive the vehicle up the test grade, stop, and hold with the service brakes.
(b) With vehicle declutched or transmission in neutral, allow the vehicle to creep very slowly under control of the service brakes. Slowly apply the parking brake and, as the parking brake energizes, release the service brake.
(c) Repeat all steps in the down grade direction.

3. TEST CALCULATIONS AND RECORDS

3.1 Level surface.

(a) Measure and record force required to set parking brakes (lever operation).
(b) Measure and record force (line pull), that causes vehicle brake wheels to slide or rotate.
(c) Record vehicle descriptive data and test results on the test summary sheet (see Fig. 108-1).
(d) Record date, time, and place of test.

3.2 Percent grade.

(a) Measure and record force required to set parking brakes (lever operation).
(b) Record percent grade, direction of vehicle on the grade and observation of wheel roll or lock and vehicle hold or slide.
(c) Record vehicle descriptive data and test results on the test summary sheet (see Fig. 108-1).
(d) Record date, time, and place of test.

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VEHICLE: MAKE_____ MODEL_____ YEAR_____
ENGINE_____ TRANSMISSION_____ AXLE_____
LOADED WEIGHT_____ LB(kg), FRONT_____ LB(kg), REAR_____ LB(kg), TOTAL WEIGHT_____ LB(kg)
EMPTY WEIGHT_____ LB(kg), FRONT_____ LB(kg), REAR_____ LB(kg), TOTAL WEIGHT_____ LB(kg)
TIRES - SIZE_____ MAKE_____ MANUFACTURER'S DESIGNATION_____
GENERAL DATA_____
BRAKES: FRONT SIZE_____ TYPE_____ CYL DIA_____
REAR SIZE_____ TYPE_____ CYL DIA_____
LININGS - FRONT_____ REAR_____
DRUM/ROTOR TYPE - FRONT_____ REAR_____
MASTER CYL DIA_____ PEDAL RATIO_____ POWER BRAKE: YES_____ NO_____ TYPE_____
TEST INFORMATION: SPECIAL EQUIPMENT
TESTED BY_____ LOCATION_____ DATE_____

FIGURE 108-1. Test Summary Sheet

X-2751

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TEST T-109

PRECIPITATION

1. SCOPE

This test is to determine if a commodity will start and operate without damage when exposed to heavy rain and high wind.

2. TEST METHOD

2.1 Test apparatus.

2.1.1 General test apparatus.

Means to measure	Accuracy
(a) Time	Sec. 6.14 a
(b) Wind velocity	Sec. 6.16 a

2.1.2 Special test apparatus.

(a) Spray fixture.
(b) Metering device.
(c) Device for generating wind.
(d) A means to trap and collect water which may enter the exhaust system leading to the exhaust manifold.
(e) Source of water.

2.2 Test procedure.

2.2.1 Static.

(a) Park the commodity on a horizontal surface and expose to simulated rain falling toward the commodity. The volume of rain impinging on the horizontal surface shall be not less than the specified amount per square foot of wetted, horizontal area. Simulated rain shall originate from a source which produces droplets of uniform distribution and velocity similar to natural rain at the point of contact with the commodity. The rain shall be driven on and against the commodity at an angle of approximately 45 degrees to the horizontal by a generated wind of not less than 30 miles per hour.

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(b) Position the commodity to expose the exhaust opening to the most critical direction for water infiltration with respect to the wind. Shut down engine.
(c) Insert item 2.1.2(d) in the exhaust system where it will trap and collect water entering the exhaust system.
(d) Expose the commodity to the rain and wind as specified in item 2.1.2(a) for a period of 5 minutes.
(e) Stop the wind and rain.
(f) Examine item 2.1.2(d) for the presence of water.

2.2.2 Operating.

(a) Expose the top and left side of the commodity to the rain as described in item 2.1.2(a) for a period of 5 minutes with the engine shut down.
(b) With water flowing, start engine unaided within 2 minutes (without removing accumulated water) and allow engine to idle for 2 minutes.
(c) Repeat 2.2.2(a) and 2.2.2(b) with the top and front of the commodity exposed.
(d) Repeat 2.2.2(a) and 2.2.2(b) with the top and right side of the commodity exposed.
(e) Repeat 2.2.2(a) and 2.2.2(b) with the top and rear of the commodity exposed.

2.2.3 Examination. During or at the conclusion of the test, examine the commodity for the following:

(a) Water in the air-intake system.
(b) Water leakage into the torque-converter oil chamber, engine crankcase sump, transmission, final drive case, fuel tank, and hydraulic reservoir.
(c) Performance of the electrical components.
(d) Functioning of all gages and instruments.
(e) Water in the dry compartments.

3. TEST CALCULATIONS AND RECORDS

Record the following:

(a) Date, time, and place of test.
(b) Commodity make, model, and serial number.
(c) Water leakage.
(d) Water content in each compartment or system.
(e) Any performance malfunction.

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TEST T-110

RAIL IMPACT

1. SCOPE

This test, commonly referred to as the Association of American Railroads (AAR) Method is used to determine if an item is capable of absorbing the impacts encountered in railroad shipment.

2. TEST METHOD

2.1 Test course. A railroad track site of sufficient distance to allow for the required acceleration and speeds shall be used.

2.2 Test apparatus.

2.2.1 General test apparatus.

Means to measure	Accuracy
(a) Speed	Sec. 6.11 b
(b) Time	Sec. 6.14 a
(c) Distance	Sec. 6.7 a

2.2.2 Special test apparatus.

(a) Five empty freight cars equipped with conventional draft gears (buffer cars).
(b) Test car (hammer car) equipped with conventional draft gears.
(c) Means of accelerating test car to specified speeds.

2.3 Test procedure.

2.3.1 Commodity placement. Load commodity on the test car and tiedown as specified.

2.3.2 Buffer car placement. The string of five empty buffer cars shall be placed with each car's airbrakes set and with the train slack removed. Buffer cars shall be repositioned after each impact. Brakes and slack shall be checked with each positioning to minimize the changes in preimpact conditions.

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2.3.3 Test car. The test car shall be accelerated to the desired velocity. If a locomotive is used to accelerate the test car, the test car shall be uncoupled from the locomotive just prior to impact and allowed to roll free over the timing instrument and impact into the string of buffer cars.

2.3.4 Impact speeds. The impacts shall be made at the specified speeds for the specified number of impacts. The test car shall then be reversed and again impacted at the specified speed. If a test speed is not attained, the impact shall be rerun.

2.3.5 Option for reversed impact. If field conditions prohibit test car reversal, the test car shall be placed at the impacted end of the string of buffer cars and impacted with a hammer car of equal or greater weight than the test car.

3. TEST CALCULATIONS AND RECORDS

Record the following:

(a) Date, time, and place of test.
(b) Commodity make, model, serial number, and weight.
(c) Method of tying commodity to test car.
(d) Any breakage or deformation.

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TEST T-111

RAIL TRANSPORTABILITY

1. SCOPE

This test is to determine whether the commodity can be transported by rail in accordance with Association of American Railroads requirements and in accordance with Berne International requirements.

2. TEST METHOD

2.1 Test apparatus.

(a) Rigid mockup in accordance with the Association of American Railroad clearance diagram (see Fig. 111-1).
(b) Rigid mockup in accordance with the Berne International clearance diagram (see Fig. 111-2).

2.2 Test procedure.

(a) Run the commodity through the Association of American Railroad clearance mockup.
(b) Run the commodity through the Berne International clearance mockup.
(c) Record any clearance problems.

3. TEST CALCULATIONS AND RECORDS

Record the following:

(a) Date, time, and place of test.
(b) Commodity make, model, serial number, and weight.
(c) Any clearance problems when the commodity runs through the mockup.
(d) Attachments or accessories removed to meet clearance requirements.

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OUTLINE DIAGRAM OF APPROVED LIMITED CLEARANCES OF ASSOCIATION OF AMERICAN RAILROADS

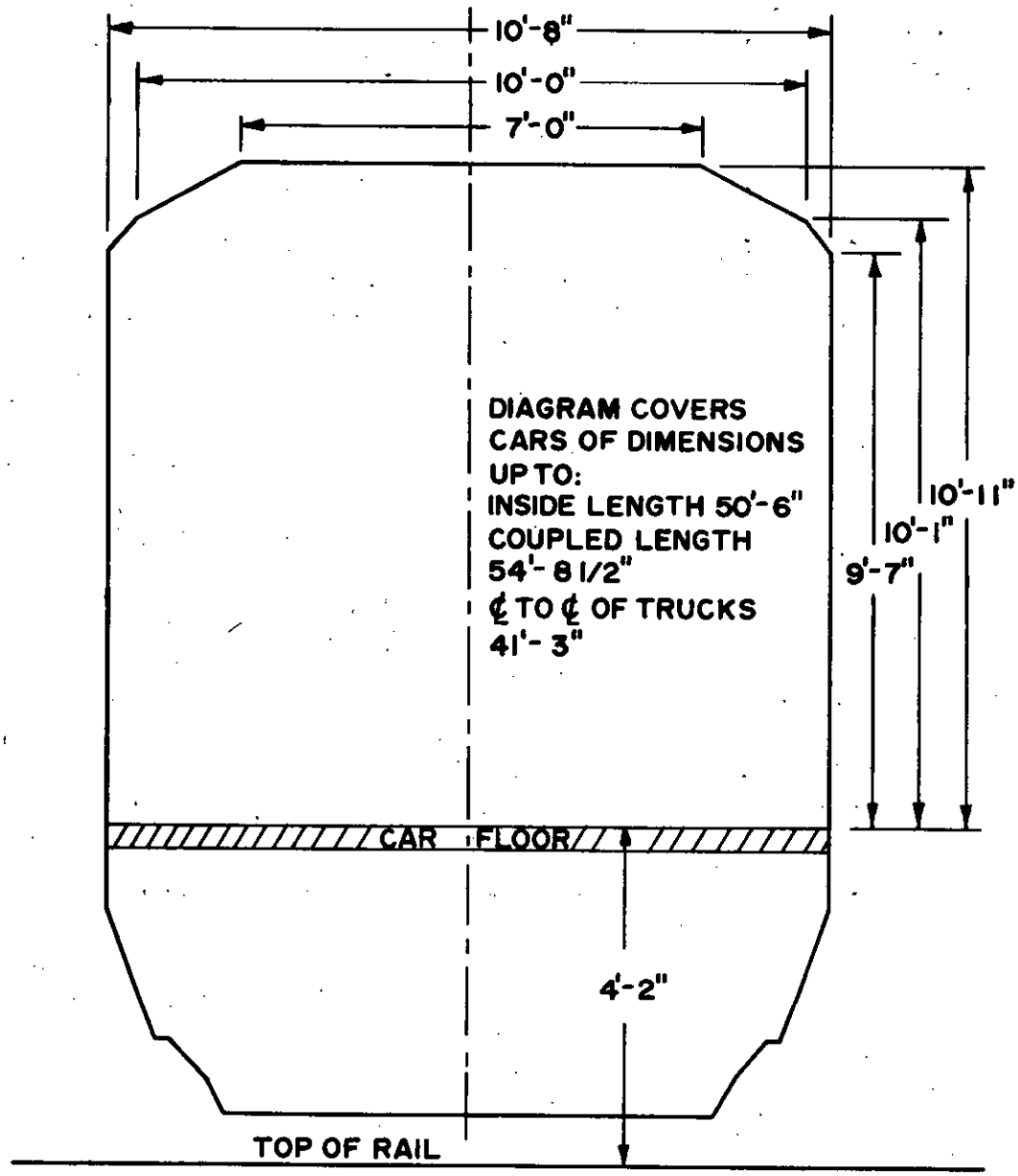
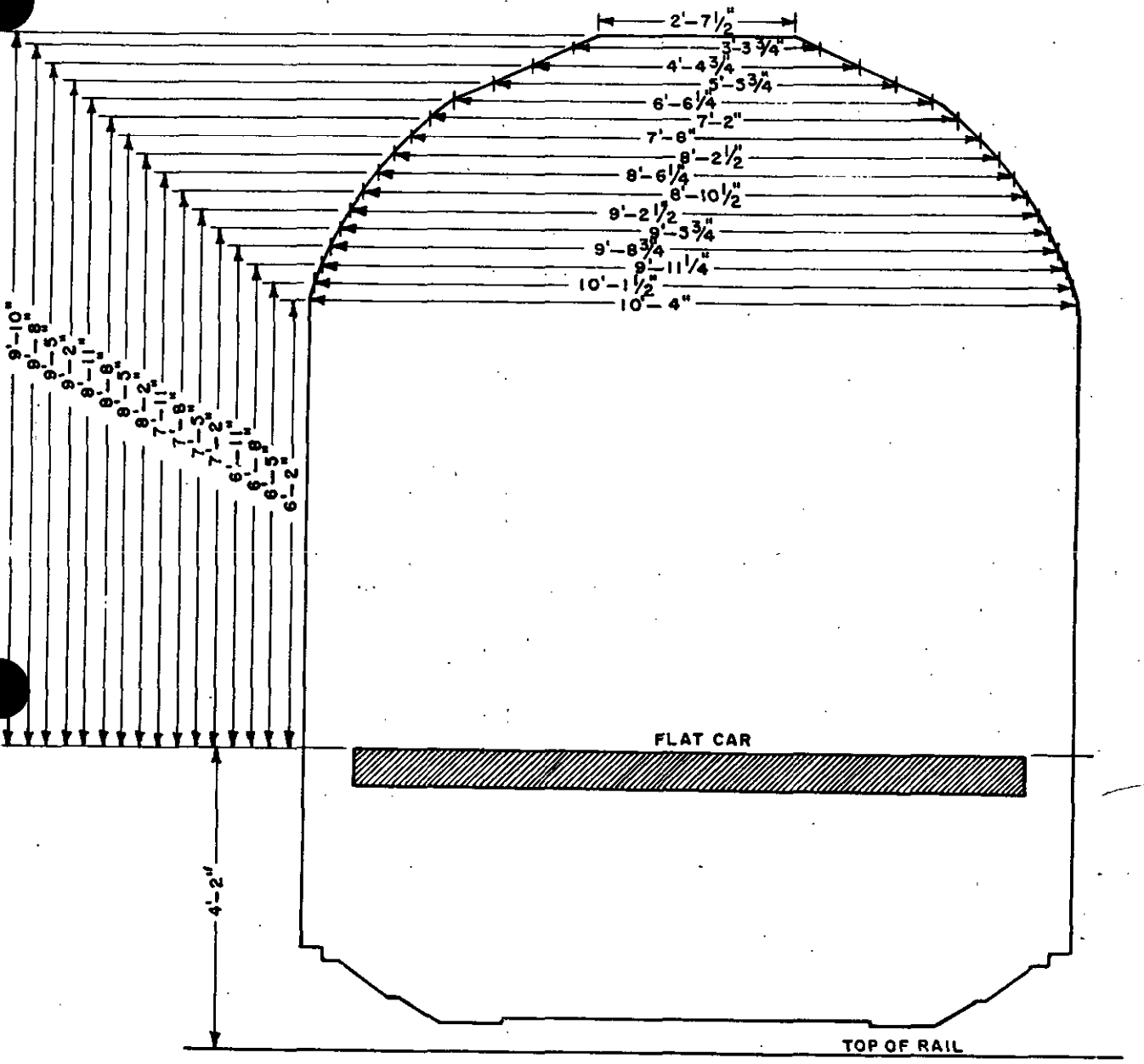


FIGURE III-1.

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ITEM LENGTH-32'-10"

FIGURE 111-2. BERNE INTERNATIONAL
(CLEARANCE DIAGRAM)

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TEST T-112

RESERVE TRACTIVE ABILITY

1. SCOPE

This test determines the ability of a self-propelled vehicle to develop reserve tractive force and power, beyond that required to move itself, on a specified course. Reserve tractive ability is reported as reserve tractive force in pounds and as reserve tractive power in horsepower.

2. TEST METHOD

2.1 Limitations.

(a) There should be a minimum of steering during the recorded runs. Revolutions of the drive wheels on wheeled vehicles shall not vary from each other by more than 3 percent. Revolutions of the drive sprockets on track vehicles shall not vary from each other by more than 2 percent.

(b) During any recorded run the instantaneous speed of engine or of the output shaft of continuously variable drive, if vehicle is so equipped, should not vary more than ± 3 percent from the specified speed. To avoid loss of power through engine overspeeding into the range of governor regulation, the governor may be set for higher engine speed or the engine may be controlled manually during the recorded run.

2.2 Test apparatus.

2.2.1 General test apparatus.

Means to measure	Accuracy
(a) Time	Sec. 6.14 a
(b) Distance	Sec. 6.7 a
(c) Engine speed and revolution	Sec. 6.11 a
(d) Output shaft speed and revolution	Sec. 6.11 a
(e) Revolutions of each drive wheel or sprocket	Sec. 6.7 a
(f) Vehicle weight	Sec. 6.15 a
(g) Tire pressure	Sec. 6.9 b
(h) Temperature	Sec. 6.13 a
(i) Barometric pressure	Sec. 6.9 a
(j) Reserved tractive force	Sec. 6.5 a

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2.2.2 Special test apparatus. Suitable towed load controllable to maintain within specified limits either:

(a) Speed of engine or of output shaft of continuously variable drive if vehicle is so equipped.
(b) Reserve tractive force.

2.3 Test course. The surfaces of the test course shall be prepared to provide desired conditions of traction and vehicle drag, see applicable provisions in Part I, Sections 5.2 and 5.3 of this standard. It may be a loop or a straight course with sufficient room for the test train to turn at the ends. In either case, it shall include a straight test zone with a minimum length (in feet) of 44 times the expected test speed (in mph) or 100 feet, whichever is greater plus sufficient length to permit stabilization of operation before start of the recorded run.

2.4 Preparation of equipment. Install test apparatus and operate sufficiently to determine that all function properly.

2.5 Test procedure.

(a) Determine the distance the vehicle travels per revolution of the drive wheels or sprockets without tractive load by either driving the vehicle in lowest gear with engine throttled to its lowest uniform running speed and without directional correction for a distance not less than 100 feet. Count revolutions of drive wheels or sprockets; divide distance traveled by revolutions to obtain travel per revolution; or if vehicle drag does not exceed 100 pounds per ton, by towing or pushing the vehicle at the lowest practical uniform speed to determine the distance traveled per revolution of the drive wheels or sprockets.
(b) Attach towed load to the vehicle in such a manner that the normal operation will be duplicated as nearly as possible. If the vehicle is designed for towing, attachment should be at the towing hitch; otherwise, it should be as close to the ground as practicable. The line of draft should be horizontal with the longitudinal axis of the vehicle unless such arrangement is impossible because some particular function dictates otherwise.
(c) Record general data required by test log (see Fig. 112-1).

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(d) Record, while traveling over the test course with towed load adjusted to maintain the average speed of drive wheels or sprockets at specified rpm.
  (1) The reserve tractive force.
  (2) Time.
  (3) Distance.
  (4) Number of revolutions of engine and of output shaft of continuously variable drive, if vehicle is so equipped.
  (5) Number of revolutions of each drive wheel or sprocket.

(e) Recorded test runs shall cover a minimum of 0.5 minutes or a minimum distance of 100 feet, whichever is greater, the average of four runs (two best runs in each direction) shall be used in reporting vehicle performance at each selected speed or load.

2.5.1 Alternate procedure.

(a) Control reserve tractive force and hold as constant as possible for each run.

(b) Record data as required in 2.5(d). The instantaneous engine and vehicle ground speed values at the end of the test run shall be within 3 percent of those at the start.

2.5.2 In order to obtain satisfactory readings in low gear on vehicles with sufficient power to slip their wheels or tracks, it may be necessary to reduce the draft load so that travel efficiency is not less than 85 percent for wheel vehicles and 93 percent for track vehicles.

3. TEST CALCULATIONS AND RECORDS

3.1 General.

(a) The above procedures show vehicle performance under a specific condition. When a broader investigation is desired, a series of recorded runs may be made in any transmission gear under several conditions of speed or reserve tractive force. This data may be plotted to show general performance characteristics on the selected course.

(b) Reserve tractive force and reserve tractive power are reported as observed, since the undetermined factors of engine friction, chassis drive train friction and vehicle drag make engine correction formulas for atmospheric conditions inapplicable; thus ambient temperature, barometric pressure, and humidity (water vapor pressure) for each test, however, are recorded on the summary sheet to show comparability of conditions with those of similar tests.

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3.2 Calculations.

$$\text{Travel efficiency, \%} = \frac{s_o \times 100}{s_s}$$

$$\text{Vehicle speed, mph} = \frac{s}{88\ t}$$

$$\text{Reserve Tractive Horsepower } (THP_o) = \frac{F_o \times \text{mph}}{375}$$

$$\text{Average speed of engine, rpm } (N_e) = \frac{N_e}{t}$$

Average speed of output shaft of continuously variable drive,

$$\text{rpm } (N_o) = \frac{N_o}{t}$$

F_o = Reserve tractive force, pounds averaged for either time or distance.

t = Time to travel test distance, minutes.

N_e = Total revolutions of engine to travel test distance.

N_o = Revolutions of output shaft of continuously variable drive to travel test distance.

N_w = Average revolutions of drive wheels or sprockets to travel test distance.

s = Test distance traveled, feet.

s_s = Average distance vehicle traveled without tractive load per revolution of the drive wheels or sprockets, feet.

s_o = Average distance vehicle traveled with tractive load per revolution of the drive wheels or sprockets, feet = $\frac{s}{N_w}$

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RESERVE TRACTIVE ABILITY TEST SUMMARY

GENERAL DATA

VEHICLE: Make_____ Model_____ Serial No._____

VEHICLE ATTACHMENT: Type_____

Make_____ Model_____ Serial No._____

ENGINE: Make_____ Model_____ Serial No._____

Rated rpm_____ Fuel_____ Fuel sp gr_____ at_____ °F

TORQUE CONVERTER: Make_____ Model_____ Serial No._____

TEST COURSE: Material_____ Condition_____

Location_____

PAYLOAD OR BALLAST: Description_____

Weight_____ lb Location_____

WHEEL OR TRACK: Load_____ Location_____

TIRE OR TRACK: Type_____ Size_____

TIRE: Rim Width_____ Inflation Pressure_____

GROSS VEHICLE WEIGHT_____ lb

HITCH HEIGHT ABOVE GROUND_____ in.

AVERAGE DATA ON INDIVIDUAL TEST

TEST NUMBER		1	2	3	4	
Log Sheet No.						
Transmission Gear						
Over-all Ratio						
Engine Speed (rpm)	n_e					
OUTPUT Shaft (rpm)	n_o					
Ground Speed	mph					
Reserve Tractive Force	F_o					
Travel Efficiency *	S					
Reserve Tractive HP	THP_o					
Ambient Temperature	F					
Barometer	In Hg					
Relative Humidity	%					

*State whether vehicle was driven ( ), towed ( ), or pushed ( ) to obtain distances traveled per revolution of drive wheel or sprocket.

REMARKS:_____

_____ DATE

_____

_____ SIGNED

FIGURE 112-1. Test Summary Sheet

X-2753

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TEST T-113

SERVICE BRAKE EFFECTIVENESS

1. SCOPE

This test determines the effectiveness of the service brake in stopping the vehicle from a specified speed.

2. TEST METHOD

2.1 Test apparatus.

2.1.1 General test apparatus.

Means to measure		Accuracy
(a)	Tire pressure	Sec. 6.9 b
(b)	Vehicle weight	Sec. 6.15 a
(c)	Wind velocity	Sec. 6.16 a
(d)	Temperature, ambient	Sec. 6.13 a
(e)	Time	Sec. 6.14 a
(f)	Vehicle speed	Sec. 6.11 b
(g)	Braking distance	Sec. 6.7

2.1.2 Special test apparatus.

(a) Electrical detonator.
(b) Odometer.

2.2 Test course. The test course shall be a clean swept, dry, brushed concrete surface with a grade not more than ± 1 percent with sufficient length to perform the test.

2.3 Test procedure.

(a) Attach the brake apparatus* for measuring braking distance so that measurement begins during the first 1/4 inch of brake pedal travel.
(b) With the vehicle traveling forward on the test course, obtain specified speed and with the engine disengaged or as specified from the drive train, apply brake pedal input force until vehicle stops.
(c) Make at least two test runs to assure repeatability.

* Electrical detonator, odometer or other means.

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3. TEST CALCULATIONS AND RECORDS

(a) Record brake pedal input force prior to test.
(b) Record vehicle speed before braking.
(c) Record the braking distance.
(d) Record vehicle descriptive data and test results on the test summary sheet (see Fig. 113-1).
(e) Record date, time, and place of test.

Note. Record any occurrence of wheel slide, loss of directional control or vehicle instability.

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VEHICLE: MAKE _____ MODEL _____ YEAR _____
ENGINE _____ TRANSMISSION _____ AXLE _____
LOADED WEIGHT _____ LB(kg), FRONT _____ LB(kg), REAR _____ LB(kg), TOTAL WEIGHT _____ LB(kg)
EMPTY WEIGHT _____ LB(kg), FRONT _____ LB(kg), REAR _____ LB(kg), TOTAL WEIGHT _____ LB(kg)
TIRES - SIZE _____ MAKE _____ MANUFACTURER'S DESIGNATION _____
GENERAL DATA _____
BRAKES: FRONT SIZE _____ TYPE _____ CYL DIA _____
REAR SIZE _____ TYPE _____ CYL DIA _____
LININGS - FRONT _____ REAR _____
DRUM/ROTOR TYPE - FRONT _____ REAR _____
MASTER CYL DIA _____ PEDAL RATIO _____ POWER BRAKE: YES _____ NO _____ TYPE _____
TEST INFORMATION: SPECIAL EQUIPMENT
TESTED BY _____ LOCATION _____ DATE _____

FIGURE 113-1. Test Summary Sheet

X-2751

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TEST T-114

SLINGING PROVISIONS

1. SCOPE

This test is used to determine adequacy of slinging provisions for end items and their subassemblies.

2. TEST METHOD

2.1 Test apparatus.

2.1.1 General test apparatus.

Means to measure	Accuracy
(a) Force	Sec. 6.5 a
(b) Linear dimensions	Sec. 6.7
(c) Gradient	Sec. 6.6

2.1.2 Special test apparatus. Lifting equipment and slings.

2.2 Test procedure. Measure the provision openings and clearance dimensions. Lift the commodity and hold in suspension in its normal lifting position. Determine the distance from the sling apex to the lowest extremity of the commodity when suspended. Determine the forward angles of application for each provision. Measure the minimum clearance between each sling and the equipment. Lower the commodity and apply the specified forces by either of the following procedures:

(a) Restrain the commodity by anchoring the main frame and subject each provision to the specified force. This force will be in the direction as determined above.

(b) Lift the commodity and add weights to the main frame until the specified force on each provision is obtained. Hold the commodity under either of the above conditions for the specified time.

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3. TEST CALCULATIONS AND RECORDS

Record the following:

(a) Date, time, and place of test.
(b) Commodity make, model, serial number, and weight.
(c) Provision clearances and dimensions.
(d) Apex height to lowest part of suspended commodity.
(e) Force, application angle, and time applied.
(f) Method of applying load to commodity.
(g) Any breakage or deformation.

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TEST T-115

TIEDOWN PROVISIONS

1. SCOPE

This test is used to determine adequacy of tiedown provisions for end items and their subassemblies.

2. TEST METHOD

2.1 Test apparatus.

2.1.1 General test apparatus.

Means of measuring	Accuracy
(a) Force	Sec. 6.5 a
(b) Gradient	Sec. 6.6
(c) Linear dimensions	Sec. 6.7

2.1.2 Special test apparatus.

(a) Means of applying a force to the tiedown provision.
(b) Means of holding the commodity in place (such as wire rope).

2.2 Test procedure. Measure the provision openings and clearance dimensions. Anchor the equipment by means other than the tiedown provisions to be tested and apply the specified forces by either of the following procedures:

(a) Apply the three directional forces as specified simultaneously or as one resultant force (composed of the three directional forces) to the provision under test. If a single resultant force is used, the angle of force application must be determined.
(b) Apply the three directional forces as specified separately (one at a time) to the provision under test. The forces shall be divided proportionally among only the tiedown provisions which are effective in resisting movement of the commodity due to the specified forces. When additional tiedowns are furnished by the supplier, a proportional force will be applied to this point(s).

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3. TEST CALCULATIONS AND RECORDS

Record the following:

(a) Date, time, and place of test.
(b) Commodity make, model, serial number, and weight.
(c) Tiedown provision clearances and dimensions.
(d) Force and time applied.
(e) Force angles if applicable.
(f) Method of holding equipment or item in place.
(g) Any breakage or deformation.

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TEST T-116

VEHICLE DRAG

1. SCOPE

This test is used to determine the external force required to maintain travel speed on a specified test course and surface. It is primarily used for determining the drag of towed construction and industrial vehicles, and their combination with mounted equipment and with or without payload. Also, it may be used to determine the resistance to motion of a self-propelled vehicle.

2. TEST METHOD

2.1 Test apparatus.

2.1.1 General test apparatus.

Means to measure		Accuracy
(a)	Gradient	Sec. 6.6 a
(b)	Time	Sec. 6.14 a
(c)	Distance	Sec. 6.7 a
(d)	Temperature, ambient	Sec. 6.13 a
(e)	Tire pressure	Sec. 6.9 b
(f)	Vehicle weight	Sec. 6.15 a
(g)	Vehicle speed, mph	Sec. 6.11 b
(h)	Towing force	Sec. 6.5 a
(i)	Wind velocity	Sec. 6.16 a

2.2 Test course. The test course shall be uniform in surface and grade as specified, (see applicable provisions in Part I, Section 5.3 of this standard) and maintained in such condition as to be consistent with the objective of the test. It may be a loop or a straight course with sufficient room for turnarounds at the ends. In either case, it shall include a straight test zone with a minimum length (in feet) of 44 times the expected test speed (in mph) or 100 feet, whichever is greater.

2.3 Preparation of equipment. Install test apparatus and operate sufficiently to determine that all function properly.

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2.3.1 Attach load cell to the towed vehicle or equipment in such a manner as to simulate the pull normal to operation of the vehicle. The line of draft should be horizontal with the longitudinal axis of the vehicle unless such arrangement is impossible because some particular function dictates otherwise. If test vehicle has a steerable front axle, a bridle hitch or some other means will be required to maintain directional control.

2.4 Test procedure.

(a) Tow vehicle over course at selected nominal constant speed and record:

(1) Drag force (pounds).
(2) Time (minutes).
(3) Distance (feet).

(b) The length (feet) of reported test runs shall be adequate to produce 0.5-minute recordings. Vehicle performance measurement shall be the average of at least two test runs in each direction.
(c) Ground speeds at start and end of run for any reported test shall not differ by more than ± 5 percent of the specified speed.
(d) Any test run in which wind velocity (in any direction) exceeds 15 mph should not be reported.

3. TEST CALCULATIONS AND RECORDS

3.1 Calculate average travel speed in mph from the time-distance values as:

$$mph = \frac{s}{88t}$$

Where
s = distance in feet.
t = time duration of test in minutes.

3.2 Record vehicle descriptive data and test results on the test summary sheet (see Fig. 116-1).

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VEHICLE DRAG TEST SUMMARY

GENERAL DATA

EQUIPMENT	MAKE	MODEL	SERIAL NUMBER
VEHICLE			
ENGINE			
VEHICLE ATTACHMENT TYPE _____			

VEHICLE WEIGHT _____ lbs

WHEEL OR TRUCKS — Type _____
Location _____ Inflation Pressure _____

LOAD CELL HITCH POINT — Height _____ in.
Location _____

TEST COURSE — Material _____ Surface _____
Location _____

PAYLOAD OR BALLAST — Weight _____
Location _____
Describe _____
_____

AVERAGE DATA ON TEST RESULTS

TEST NUMBER		ONE DIRECTION 1	ONE DIRECTION 2	ONE DIRECTION AVE	OPPOSITE DIRECTION 1	OPPOSITE DIRECTION 2	OPPOSITE DIRECTION AVE	OVERALL AVE
GROUND SPEED	mph							
VEHICLE DRAG	lbs							
AMBIENT TEMPERATURE	°F							
WIND VELOCITY	mph							
WIND DIRECTION								

REMARKS _____

_____ DATE

_____

_____ SIGNED

FIGURE 116-1. Test Summary Sheet

X-2755
Test T-116

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TEST T-117

VEHICLE TRAVEL SPEED

1. SCOPE

This test determines the travel speed of a vehicle under its own power. Travel speed is determined under specified conditions and test runs may be made in any transmission gear.

2. TEST METHOD

2.1 Limitations.

(a) The instantaneous vehicle speed should not vary more than ± 3 percent during any test run.
(b) Do not record test results when the wind velocity is greater than 10 mph.

2.2 Test apparatus.

2.2.1 General test apparatus.

Means to measure		Accuracy
(a)	Wind velocity	Sec. 6.16 a
(b)	Tire pressure	Sec. 6.9 b
(c)	Atmospheric pressure	Sec. 6.9 a
(d)	Engine, rpm	Sec. 6.11 a
(e)	Travel speed	Sec. 6.11 b
(f)	Temperature, ambient	Sec. 6.13 a
(g)	Time	Sec. 6.14 a
(f)	Distance	Sec. 6.7 a

2.3 Test course. The test course shall be a clean swept, dry, brushed concrete surface or other specified surface, with a grade not more than ± 1 percent with sufficient length to perform the test adequately.

2.4 Test procedure.

(a) Ballast the vehicle as required for test.
(b) Operate the vehicle in the specified gear to assure all parts are functioning adequately.
(c) Drive the vehicle over the test course under the specified conditions, e.g. full throttle, engine rpm, for the selected gear.
(d) Conduct a total of four runs, two in each direction.

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3. TEST CALCULATIONS AND RECORDS

(a) If applicable, calculate averate travel speed in mph from time-distance values as:

$$mph = \frac{s}{88t}$$

Where
s = distance in feet.
t = time duration of test in minutes.

(b) Record vehicle descriptive data and test results on the test summary sheet (see Fig. 117-1).

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TRAVEL SPEED TEST SUMMARY

GENERAL DATA

VEHICLE:_____ Make _____ Model _____
Serial No._____ Weight _____lbs.

OEPRATING ATTACHMENT: _____
Travel Position, describe_____
_____

TRAILED EQUIPMENT: Describe _____
_____

PROPELLING ENGINE: Make _____ Model_____
Serial No._____

TEST COURSE: Material_____ Condition_____
Location _____

TIRE INFLATION PRESSURE: _____

AVERAGE DATA ON INDIVIDUAL TESTS						
TEST NUMBER		1	2	3	4	AVERAGE
WIND VELOCITY	mph					
GEAR USED						
ENGINE SPEED	rpm					
TRAVEL SPEED	mph					
AMBIENT TEMPERATURE	°F					
BAROMETER	in. Hg					

REMARKS_____ _____
DATE

_____

_____ _____
SIGNED

_____

FIGURE 117-1. Test Summary Sheet

X-2756A
Test T-117

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TEST T-201

HYDRAULIC SYSTEM CLEANLINESS

1. SCOPE

This test is used to determine the cleanliness level of a hydraulic system fluid using a liquid automatic particle counter.

2. TEST METHOD

2.1 Limitations.

(a) The particle counter can be used above its saturation limit. However, readings taken above saturation will not be accurate and can be used only as an indication of the contamination level of the fluid being checked.

(b) When the particle count is being performed by connecting the counter directly to the vehicle hydraulic system, flow through the bypass line should be high enough to insure there is no settling of contamination. (i.e. 2 gpm minimum in 1/4" line; 20 gpm minimum in 3/4" line.)

2.2 Test apparatus.

2.2.1 General test apparatus.

Means to measure	Accuracy
(a) Liquid volume	Sec. 6.4 a
(b) Time	Sec. 6.14 a
(c) Particle contamination in fluid	Sec. 6.8

2.2.2 Special test apparatus. Liquid Automatic Particle Counter calibrated per ANSI B93.28 using "AC" Fine Test Dust. Two known sources for particle counters are:

High Accuracy Products Corp.
141 Spring Street
Claremont, CA 91711

Royco Instruments, Inc.
141 Jefferson Drive
Melno Park, CA 94025

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2.3 Test procedure. The test may be performed by attaching the counter to the system or by extracting a sample for remote analysis. When the system contains cylinders which have limited fluid exchange with the recirculating system, the lines to these cylinders should be interconnected for initial cleanup. Perform the particle count as described in either 2.3.1 or 2.3.2. If cylinder lines have been interconnected, reconnect to the cylinders, and actuate controls to get full extension and retraction of the cylinders. Perform the particle count again.

2.3.1 On-the-system analysis. Connect the counter to the system using a bypass line and branch line (see definitions and Fig. 201-1). The counter shall be connected upstream of all filter(s) used for cleanup of the system. Start system and actuate all system controls for not less than 2 minutes to insure mixing of the fluid. Verify the flow rate through the sensor and bypass line. Begin counts. Counts should be allowed to stabilize before recording counts. Example of typical run of counts:

1800 > 10u/175 > 20u
1500 > 10u/100 > 20u
1100 > 10u/47 > 20u
1091 > 10u/43 > 20u
1118 > 10u/41 > 20u

If the system is being filtered, counts may continue to decrease. The required counts shall be the average of not less than three consecutive counts.

2.3.2 Remote analysis - bottle sampling. Start the system and actuate all system controls for not less than 2 minutes to assure adequate mixing of the fluid. Extract a sample with the system operating or within 2 minutes of shutdown. If the sample is extracted from a system line, it shall be taken from a turbulent point in the system, such as an elbow, and shall be upstream of all filter(s) used for cleaning of the system. If a valve is used, it shall be opened and not less than 1 pint of fluid shall be allowed to pass through the line prior to taking a sample. The valve shall not be adjusted while the sample is being collected. If the sample is taken from the system reservoir, the sample shall be taken from the bottom one-third of the reservoir. Required counts shall be the average of not less than three consecutive counts.

2.3.2.1 Sample bottles. Sample bottles shall be clean. An acceptable cleaning method shall consist of rinsing the bottles three times with solvent that has been filtered through a membrane disc of not more than 1.0 micrometer mean pore size. Other means of bottle cleaning and cleanliness verification are detailed in ANSI B93.20.

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Notes.

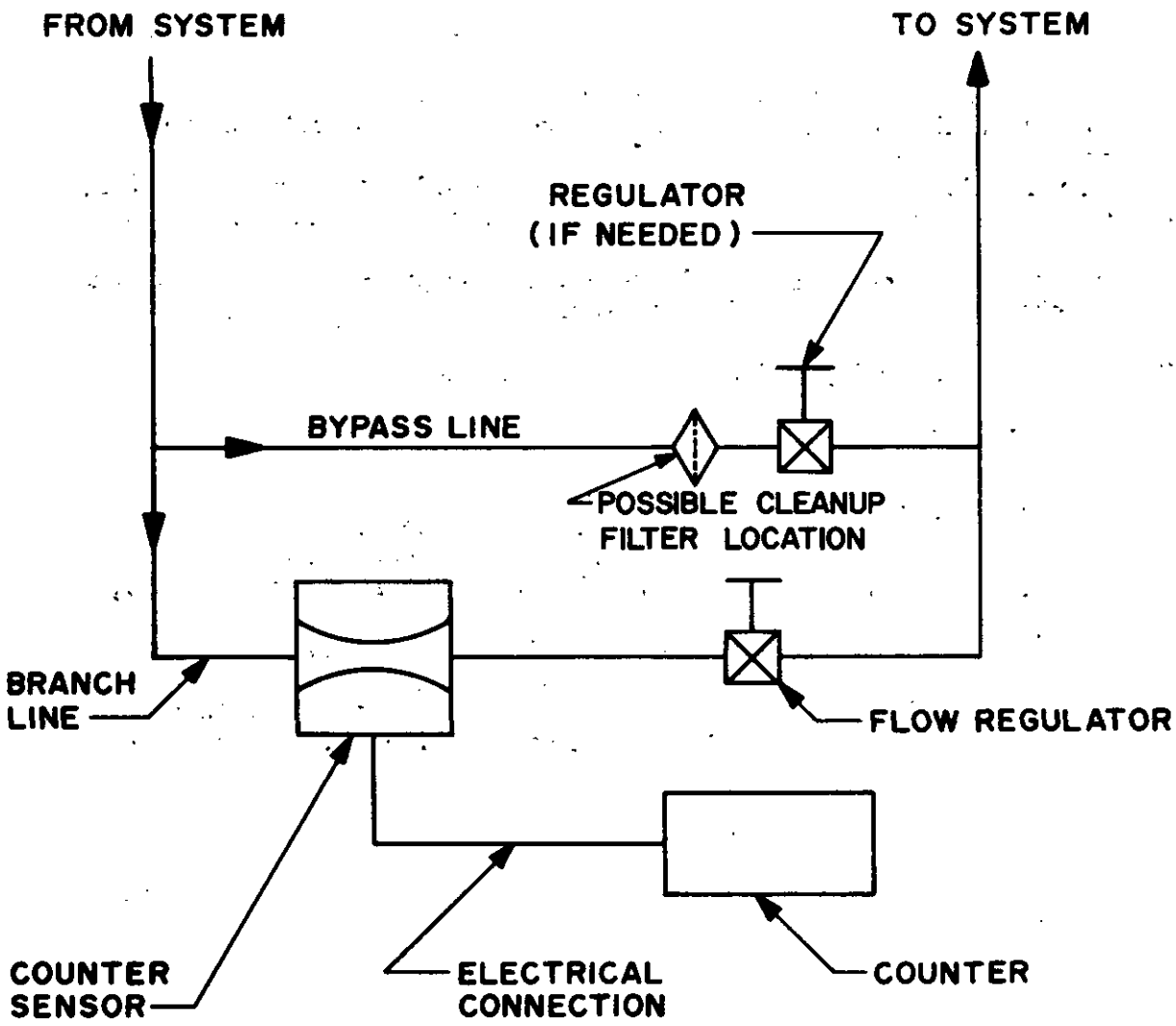
1. Branch line shall not exceed 12 inches in length.
2. Circuit shall contain provision for verifying flow rate through the counter sensor.
3. Flow regulator devices shall be downstream of the sensor (as noted).
4. Several variations of Fig. 201-1 are available as packaged units from counter manufacturers.
5. The counter manufacturer's instructions for proper operation of the counter being used should be followed.

3. TEST CALCULATIONS AND RECORDS

Record the following:

(a) Date, and location of test.
(b) Make, model, and serial number of item tested.
(c) Make, model, and serial number of counter used.
(d) Certification of calibration of counter, including data, and date of calibration.
(e) Validation of calibration prior to test, and data.
(f) Sample volume and flow rate.
(g) Whether count was run on the item or from a bottle sample.
(h) The three sets of counts that are averaged for the required count.

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ALL LINES 1/4" ID MAX.

FIGURE 201-1-TYPICAL SYSTEM HOOK-UP

X-2757A

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COMPONENT

TESTS

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TEST T-301

ACCUMULATOR

1. SCOPE

This test determines the leakage and the ability of a fluid power accumulator to withstand a proof pressure.

2. TEST METHOD

2.1 Limitations. Provide protection to prevent injury in case of item failure.

2.2 Test apparatus.

2.2.1 General test apparatus.

Means to measure	Accuracy
(a) Time	Sec. 6.14 a
(b) Pressure	Sec. 6.9 c

2.2.2 Special test apparatus. Gas cylinder and regulator. Unless otherwise specified herein, use Nitrogen gas. For oil leakage test, it may be necessary to provide external heating to maintain oil temperature for the duration of the test.

2.3 Test procedure.

2.3.1 Proof test. Completely fill both ends of the accumulator with oil and plug the gas end. Apply specified oil pressure to the oil port and maintain for not less than 60 seconds. Examine for leakage and damage.

2.3.2 Gas leakage. With the oil port open to atmosphere, pressurize the gas port with specified gas (see Fig. 301-1). Hold pressure for not less than 1 hour. Examine for leakage.

2.3.3 Oil leakage. With the accumulator mounted in the vertical position with the gas port down and open to the atmosphere (see Fig. 301-1), apply oil pressure to the oil port and hold for not less than 1 hour. Oil temperature and pressure shall be as specified in the end item specification. Examine for leakage.

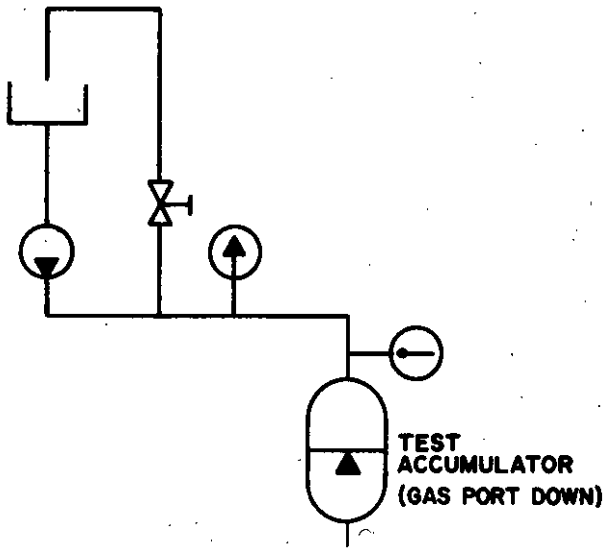
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3. TEST CALCULATIONS AND RECORDS

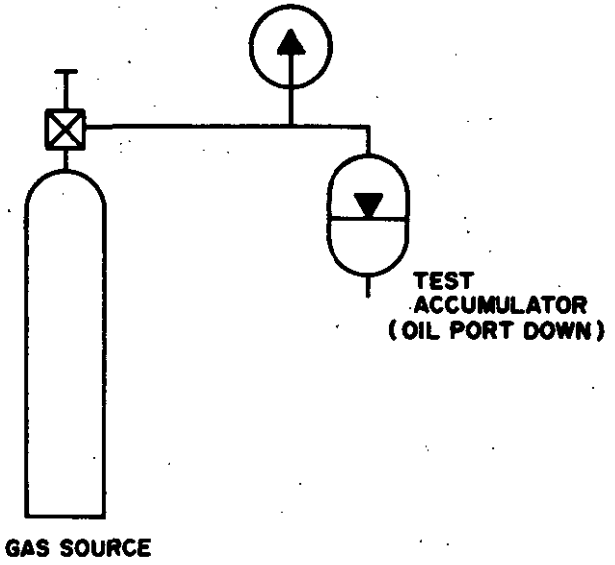
Record the following:

(a) Date, and place of test.
(b) Make, model, and serial number of accumulator tested.
(c) Gas used for test.
(d) Proof pressure used.
(e) Gas pressure used for leakage test.
(f) Oil temperature and pressure used for oil leakage test.
(g) Any visible oil leakage, either external or visible through open ports.

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PROOF & OIL LEAKAGE



GAS LEAKAGE

FIGURE 301-1. TEST HOOK-UPS

X-2758A

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TEST T-302

CYLINDER

1. SCOPE

This test is used to determine the following:

(a) Ability to withstand a proof pressure.
(b) Piston drift (internal piston leakage).
(c) Packing drag.

2. TEST METHOD

2.1 Limitations. Provide adequate protection to prevent injury in case of test item failure.

2.2 Test apparatus.

Means to measure	Accuracy
(a) Time	Sec. 6.14 a
(b) Pressure	Sec. 6.9 c
(c) Temperature	Sec. 6.13 a
(d) Distance (Linear)	Sec. 6.7 a

2.3 Test procedure.

(a) Proof pressure. Install the cylinder in the test stand. Position and mechanically hold the piston at the approximate midpoint of the cylinder. Fill both sides of the piston with oil. Establish the test temperature and pressure as specified. With the head end port capped, apply an oil pressure to the rod end of the piston for not less than 60 seconds. With the rod end port capped, apply an oil pressure to the head end of the piston for not less than 60 seconds. Examine for leakage and damage.

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(b) Piston drift. Install the cylinder in the test system. Position and hold the piston of the cylinder at the approximate midpoint of the cylinder. Fill both sides of cylinder with oil. Establish the test temperature and pressure. Cap the head end port and vent the rod end port to atmosphere. Pressurize the head end of the slave cylinder to attain a minimum of specified operating pressure at the head end of the test cylinder. Maintain this pressure for 15 minutes. Measure the travel of the piston during this period. Repeat the above test capping rod end port venting the head end port to atmosphere and pressurizing the rod end of the slave cylinder.

(c) Piston and rod packing drag (packing drag). Install the cylinder in the test system. Position the piston at the approximate midpoint of the cylinder. Fill both sides of the cylinder with oil. Establish the test temperature. Vent the head end to atmosphere and gradually pressurize the rod end of the cylinder. Record the minimum pressure at which the piston moves and also the pressure required to keep it in motion. Repeat the above test by pressurizing the head end of the cylinder.

3. TEST CALCULATIONS AND RECORDS

Record the following:

(a) Date, and place of test.
(b) Cylinder manufacturer's name, part number, and serial number (if any).
(c) Test proof pressure used external leakage and damage to cylinder.
(d) Temperature and fluid used for tests.
(e) Travel or drift of piston.
(f) Initial minimum pressure for piston travel.
(g) Minimum pressure for continuous piston travel.

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TEST T-303

DIRECTIONAL CONTROL VALVE

1. SCOPE

This test determines the following:

(a) Ability to withstand a proof pressure.
(b) Pressure drop across the valve.
(c) Internal leakage across spools.

2. TEST METHOD

2.1 Limitations. Provide protection to prevent injury of personnel in case of test item failure.

2.2 Test apparatus.

Means to measure	Accuracy
(a) Time	Sec. 6.14 a
(b) Pressure	Sec. 6.9 c
(c) Temperature	Sec. 6.13 a
(d) Flow	Sec. 6.4

2.3 Test procedure.

(a) Proof pressure test. Install the control valve in the test system. Establish the test temperature and pressure as specified. With the control valve spool in the neutral position and return port open, pressurize the cylinder ports and pressure posts simultaneously to establish pressure at a pressure rise rate of not less than 25,000 psi per second. Hold the pressure for not less than 60 seconds.

(b) Pressure drop. Install the control valve in the test stand. Establish the test temperature and rated flow. Cycle the valve through its operating functions and record the pressure drop across the valve.

(c) Internal leakage. Install the control valve in the test stand. Establish the test temperature, flow, and pressure. With the spool in neutral position, pressurize each cylinder port and measure the leakage at tank port for each spool.

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3. TEST CALCULATIONS AND RECORDS

Record the following:

(a) Date, and place of test.
(b) Control valve manufacturer's name, part number, and serial number (if any).
(c) Test proof pressure used.
(d) Temperature and fluid used for tests.
(e) Flow used for tests.
(f) Any external leaks at sealing surfaces.
(g) Pressure used for pressure drop and leakage tests.
(h) Internal leakage at each spool measured.

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TEST T-304

FILTER ASSEMBLY

1. SCOPE

This test is used to determine values for the following parameters for a fluid filter:

(a) Housing proof pressure.
(b) Clean filter assembly pressure drop.
(c) Bypass valve opening pressure.
(d) Operation of the filter assembly bypass valve indicator.
(e) Pressure drop across a bypass valve.

2. TEST METHOD

2.1 Limitations. When performing proof pressure test, increase pressure slowly and provide suitable protection to prevent injury to personnel in the event of housing failure.

2.2 Test apparatus.

Means to measure	Accuracy
(a) Pressure	Sec. 6.9 c
(b) Time	Sec. 6.14
(c) Flow	Sec. 6.4 a
(d) Temperature	Sec. 6.13
(e) Liquid volume	Sec. 6.4 a

2.3 Test procedure.

2.3.1 Housing proof pressure (Test Circuit 1, 2, or 3).

(a) Block housing outlet port. Use either power-driven or hand-operated pump.
(b) Pressurize the filter housing to the specified pressure and hold for not less than 60 seconds.
(c) Examine for leaks.

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2.3.2 Assembly pressure drop (Test Circuit 1). Install the filter with a new element in the test stand or on the vehicle. Establish maximum system flow through the filter at the oil temperature specified. Measure the pressure drop across the filter assembly.

2.3.3 Bypass valve opening pressure and operation of indicator (Test Circuit 2 or 3) (see Fig. 304-1). Plug the filter element chamber oil exit in the housing or otherwise obstruct all flow through the filter element. This may be done with a suitable rubber plug or metal plug with appropriate gaskets. Reinstall the housing in the test circuit with outlet port open. Use either a power-driven or hand-operated pump. Increase upstream pressure until the indicator enters the bypass area or bypass valve leakage exceeds 5 percent of maximum system flow, whichever occurs first. Observe pressure and indicator flow.

2.3.4 Bypass valve pressure drop (Test Circuit 1) (see Fig. 304-1). Plug the filter element chamber oil exit in the housing or stop flow through the element as described in 2.3.3. Install the housing in the test circuit. Establish maximum system flow through the filter at the oil temperature specified. Measure the pressure drop across the bypass valve.

Note. Bypass valve pressure drop can be assumed to be the pressure drop across the housing measured with the flow through the bypass valve minus the filter housing pressure drop. The housing pressure drop can be determined using the assembly pressure drop procedure (see 2.3.2), run with no element in the housing.

3. TEST CALCULATIONS AND RECORDS

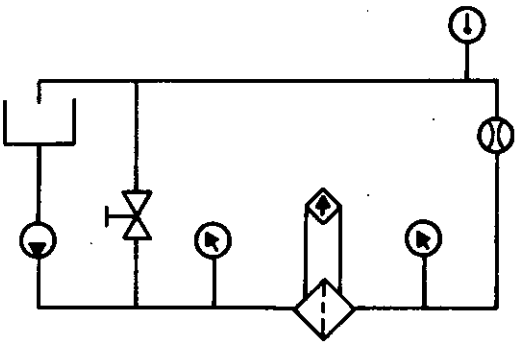
Record the following data:

(a) Date, and location of test.
(b) Location of filter in the system (i.e. pressure line, suction line, return line).
(c) Manufacturer's designation of test oil used.
(d) Make and model of filter housing.
(e) Make and model of filter element.
(f) Any external filter housing leaks.
(g) Pressure drop across filter assembly.
(h) Pressure at which bypass indicator enters "bypass" area or bypass valve leakage exceeds 5 percent of maximum system flow.

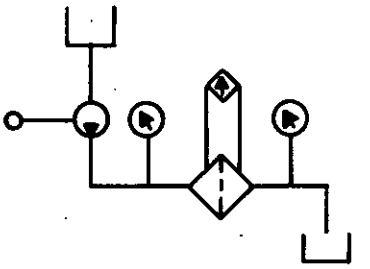
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TYPICAL TEST SCHEMATIC

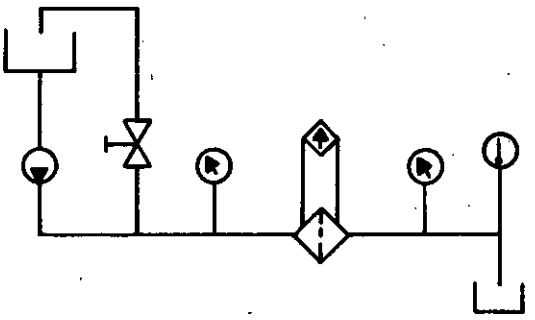
(TEST CIRCUIT 1)



(TEST CIRCUIT 3)



(TEST CIRCUIT 2)



CALIBRATED CONTAINER

SEE PART I, SECTION 2 FOR LIST OF SYMBOLS.

FIGURE 304-1. TEST CIRCUITS

X-2759A

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TEST T-305

HOLDING VALVE

1. SCOPE

This test determines the ability of a fluid power holding valve to withstand a proof pressure.

2. TEST METHOD

2.1 Limitations. Provide shielding of the test components to prevent injury to personnel in case of test item failure.

2.2 Test apparatus.

Means to measure	Accuracy
(a) Time	Sec. 6.14
(b) Pressure	Sec. 6.9 c

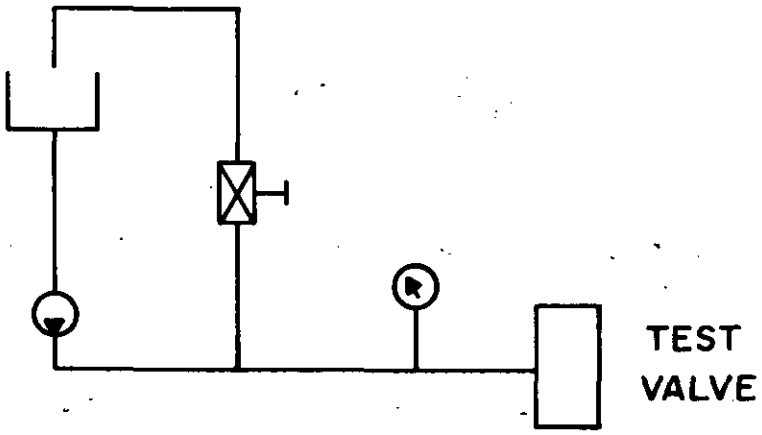
2.3 Test procedure. Pressurize one port of the holding valve to the required pressure and hold for not less than 60 seconds (see Fig. 305-1). Repeat with each of the other ports. Examine for leakage.

3. TEST CALCULATIONS AND RECORDS

Record the following:

(a) Date, and location of test.
(b) Test fluid used.
(c) Make and model of valve tested.
(d) Test proof pressure.
(e) Any leakage or damage noted.

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EITHER POWER DRIVEN OR HAND PUMP MAY BE USED.

SEE PART I, SECTION 2 FOR LIST OF SYMBOLS

FIGURE 305-1. TYPICAL CIRCUIT

X-2760A

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TEST T-306

HOSE ASSEMBLY

1. SCOPE

This test determines the proof pressure capability of a fluid power hose assembly.

2. TEST METHOD

2.1 Limitations. Provide suitable shielding of the test item to prevent injury to personnel in case of test item failure.

2.2 Test apparatus.

Means to measure	Accuracy
(a) Pressure	Sec. 6.9 c
(b) Time	Sec. 6.14
(c) Temperature	Sec. 6.13

2.3 Test procedure. Pressurize the hose assembly to the specified proof pressure. When the specified pressure is reached, close off the hose assembly from the pressure source, and hold the pressure for not less than 60 seconds (see Fig. 306-1).

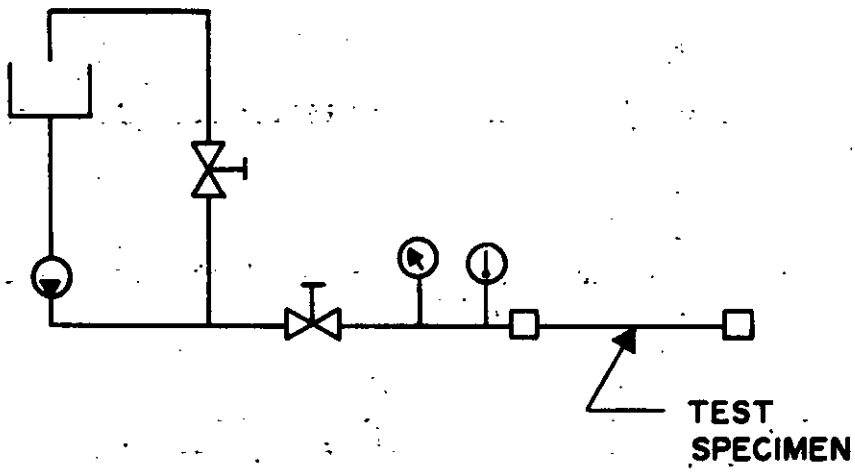
3. TEST CALCULATIONS AND RECORDS

Record the following:

(a) Date, and location of test.
(b) Part numbers and manufacturers of hose and coupling.
(c) Test pressure and fluid temperature.
(d) Any leakage, rupture noted or detachment or slippage of coupling.

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TYPICAL TEST SETUP



SEE PART I, SECTION 2 FOR LIST OF SYMBOLS

FIGURE 306-1. TYPICAL TEST SETUP

X-2761A

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TEST T-307

PUMP/MOTOR

1. SCOPE

This test determines the ability to withstand a proof pressure and the overall efficiency of a fluid power pump or motor.

2. TEST METHOD

2.1 Limitations. Provide adequate protection to prevent injury to personnel in case of test item failure.

2.2 Test apparatus.

Means to measure	Accuracy
(a) Time	Sec. 6.14
(b) Pressure	Sec. 6.9 c
(c) Shaft speed (rpm)	Sec. 6.11
(d) Torque	Sec. 6.12 b
(e) Temperature	Sec. 6.13

2.3 Proof pressure. Install the pump/motor in the test stand. Establish test temperature, speed, and pressure. Operate the pump/motor at established temperature, speed, and pressure; hold for not less than 60 seconds (see Fig. 307-1).

2.4 Efficiency. Install the pump/motor in the test system. Establish test speed and fluid temperature as specified. Measure flow and torque at a minimum of pressure points at equal increments of established speed and temperature. Calculate overall efficiency using following formula:

$$\text{Overall efficiency } \% = \frac{\text{Output HP X 100}}{\text{Input HP}}$$

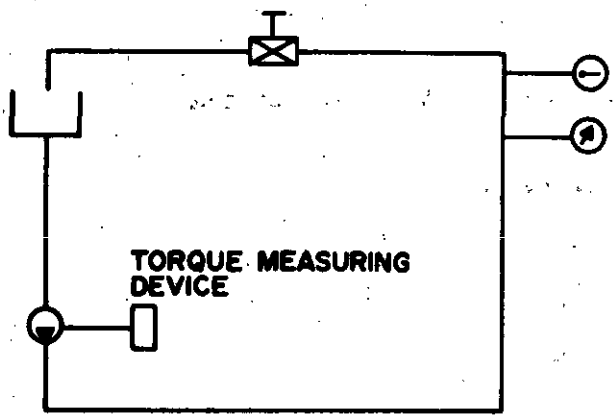
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3. TEST CALCULATIONS AND RECORDS

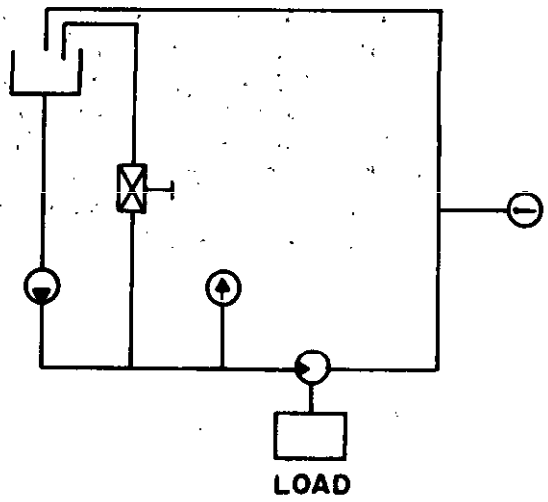
Record the following:

(a) Date, and place of test.
(b) Pump/motor manufacturer's name, part number, and serial number (if furnished).
(c) Test proof pressure used.
(d) Temperatures and fluid used for tests.
(e) Test pump/motor speeds used.
(f) Any external leaks at assembled surfaces and shaft seals.
(g) Pressures used for efficiency test. Efficiencies calculated.

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PUMP TEST



MOTOR TEST

SEE PART I, SECTION 2 FOR LIST OF SYMBOLS

FIGURE 307-1. TEST SETUPS

X-2762A

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TEST T-308

RELIEF VALVE

1. SCOPE

This test determines the following:

(a) Cracking pressure.
(b) Reseat pressure.
(c) Ability to withstand proof pressure for a fluid power relief valve.

2. TEST METHOD

2.1 Limitations.

(a) Provide protection to prevent injury to personnel in case of test item failure.
(b) Observe manufacturer's limitation in pressurizing tank ports.

2.2 Test apparatus.

Means to measure	Accuracy
(a) Time	Sec. 6.14
(b) Temperature	Sec. 6.13
(c) Flow	Sec. 6.4
(d) Liquid volume	Sec. 6.4
(e) Pressure	Sec. 6.9 c

2.3 Test procedure.

2.3.1 Proof pressure. Pressurize all pressure ports of the valve to the required pressure and hold for not less than 60 seconds. Pressure may be applied to one port and remaining pressure ports may be plugged. "Tank" port shall be left open to atmosphere.

2.3.2 Cracking pressure. Assemble valve in the test stand with tank port open and facing downward. With designated fluid in the test stand, establish test temperature. Slowly apply increasing pressure to the valve pressure port until not more than 0.1 gpm flows from tank port.

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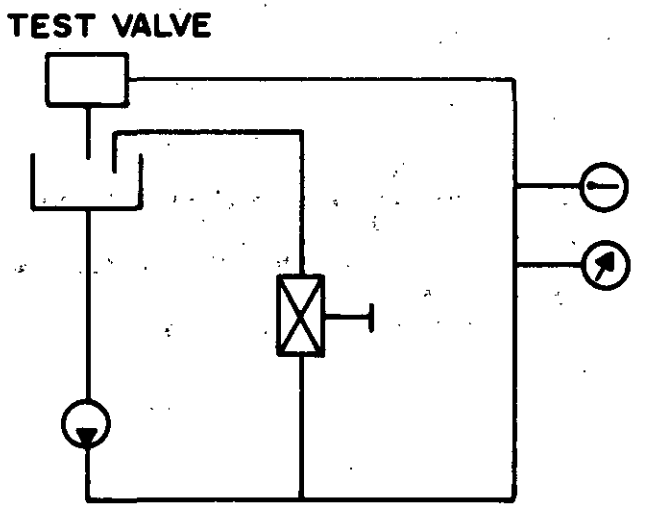
2.3.3 Reseat pressure. Assemble the valve in the test stand with the tank port open and facing downward. A hose may be connected to the tank port to direct the flow into the reservoir. With the designated fluid in the test stand, establish test temperature. Increase pressure and flow at the pressure port until the flow from the tank port reaches maximum valve rated flow. Slowly decrease pressure at the pressure port until the valve reseats sufficiently to limit flow through the tank to not more than 0.1 gpm.

3. TEST CALCULATIONS AND RECORDS

Record the following:

(a) Date, and place of test.
(b) Test valve manufacturer's name and part number.
(c) Test proof pressure used.
(d) Temperature and fluid flow used for cracking and reseat test and pressures observed.
(e) Any damage or leakage during proof test.

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FIGURE 308-1. TYPICAL TEST CIRCUIT

X-2763A

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TEST T-309

RESERVOIR

1. SCOPE

This test determines the proof pressure capability of a hydraulic reservoir.

2. TEST METHOD

2.1 Limitations. Avoid overpressurization of the reservoir. Provide adequate protection to prevent injury to personnel in case of item failure.

2.2 Test apparatus.

Means to measure	Accuracy
(a) Time	Sec. 6.14
(b) Pressure	Sec. 6.9 c

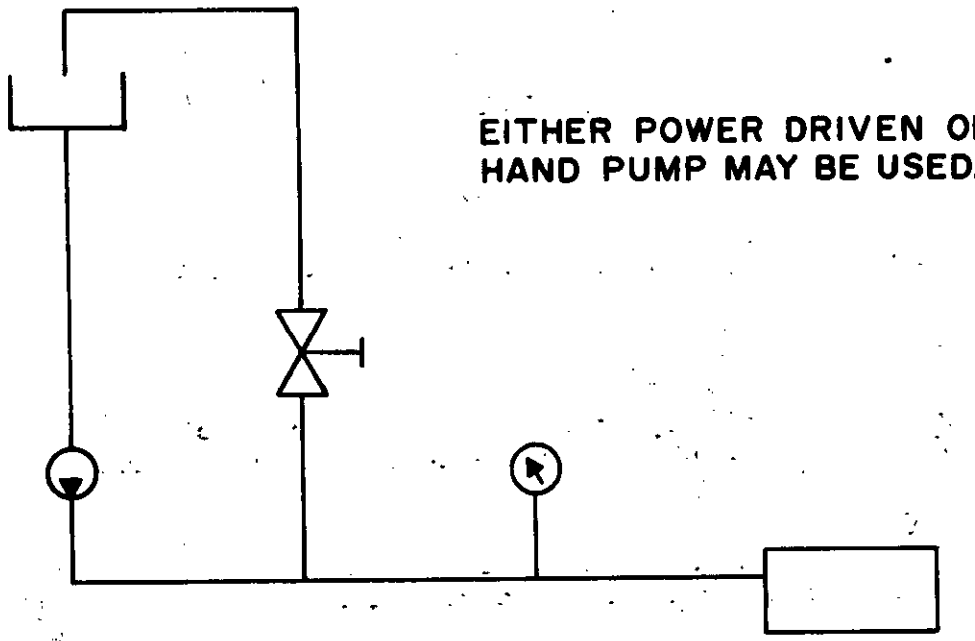
2.3 Test procedure. Fill the reservoir completely with oil. Close all outlets and pressurize to the required pressure. Hold for not less than 60 seconds. Examine for leakage.

3. TEST CALCULATIONS AND RECORDS

Record the following:

(a) Date, and location of test.
(b) Test fluid used.
(c) Make and model of reservoir.
(d) Test proof pressure.
(e) Any leakage or damage noted in reservoir.

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FIGURE 309-1. TYPICAL TEST CIRCUIT

X-2764A

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